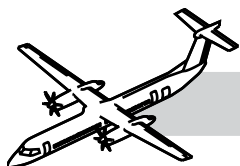


CHAPTER 5-12

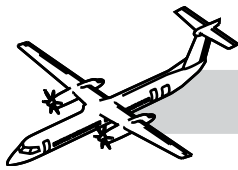
AIRCRAFT GENERAL

CONTENTS

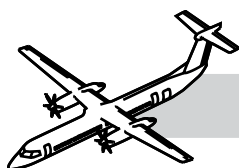
	Page
05-00-00 TIME LIMITS/MAINTENANCE CHECKS	5-1
Part 1 - Maintenance Review Board (MRB) Report.....	5-2
Revision Process	5-2
Maintenance Program.....	5-2
Part 2 - Airworthiness Limitation Items (ALI).....	5-3
Unscheduled Maintenance Checks	5-3
06-00-00 DIMENSIONS AND AREAS	6-1
06-10-00 DIMENSIONS	6-3
General	6-3
Overall Dimensions.....	6-3
06-20-00 AIRCRAFT STATIONS	6-3
General	6-3
06-30-00 AIRCRAFT ZONES.....	6-4
General	6-4
06-41-00/06-43-00 EXTERNAL & INTERNAL ACCESS PANELS AND DOORS	6-8
General	6-8
07-00-00 LIFTING AND SHORING.....	7-1
07-00-00 LIFTING AND SHORING.....	7-2
Introduction	7-2
General	7-2
07-00-00 SPECIAL TOOL & TEST EQUIPMENT.....	7-5



	Page
07-00-00 CAUTIONS & WARNINGS	7-5
08-00-00 LEVELING AND WEIGHING	8-1
08-10-00 WEIGHING OF THE AIRCRAFT	8-3
Introduction	8-3
General	8-3
08-20-00 LEVELING OF THE AIRCRAFT	8-5
Introduction	8-5
General	8-5
08-00-00 SPECIAL TOOL & TEST EQUIPMENT	8-6
08-00-00 CAUTIONS & WARNINGS	8-6
09-00-00 TOWING AND TAXIING	9-1
09-10-00 TOWING OF THE AIRCRAFT	9-3
Introduction	9-3
General	9-3
09-20-00 TAXIING OF THE AIRCRAFT	9-9
Introduction	9-9
General	9-9
09-00-00 SPECIAL TOOL & TEST EQUIPMENT	9-11
09-00-00 CAUTIONS & WARNINGS	9-11
10-00-00 PARKING, MOORING AND RETURN TO SERVICE	10-1
10-11-00 PARKING	10-2
Introduction	10-2
General	10-2
10-12-00 AIRCRAFT STORAGE	10-6
Introduction	10-6



	Page
General	10-6
Terms and References	10-6
Humidity Indicators	10-6
Engine Preservation	10-6
Environmental Considerations.....	10-6
Storage Records	10-7
10-13-00 ENGINE PRESERVATION/DEPRESERVATION	10-8
Introduction	10-8
General	10-8
10-20-00 MOORING OF THE AIRCRAFT.....	10-10
Introduction	10-10
General	10-11
10-00-00 SPECIAL TOOL & TEST EQUIPMENT.....	10-13
10-00-00 CAUTIONS & WARNINGS.....	10-14
11-00-00 PLACARDS AND MARKINGS.....	11-1
11-00-00 PLACARDS AND MARKINGS.....	11-2
Introduction	11-2
General	11-2



ILLUSTRATIONS

Figure	Title	Page
6-1	Aircraft Axis.....	6-2
6-2	Major Zones	6-4
6-3	Major Zone 100 (Lower Fuselage) Sub-Zones	6-5
6-4	Major Zone 200 (Upper Fuselage) Sub-Zones.....	6-6
6-5	Major Zone 300 (Empennage) Sub-Zone	6-7
6-6	Major Zone 400 (Nacelles) Sub-Zones.....	6-7
6-7	Access Panels and Doors	6-8
7-1	Jacking of the Aircraft	7-3
7-2	Jacking of the Aircraft at the NLG Axle.....	7-4
7-3	Jacking of the Aircraft at the MLG Axle	7-4
8-1	Weighing of the Aircraft	8-2
8-2	Method to Determine the Center Line of Main Wheels	8-2
8-3	Leveling of the Aircraft	8-4
9-1	Towing the Aircraft with a Towbar.....	9-2
9-2	Towing/Moving Aircraft with Flat Tires	9-4
9-3	Towing of the Aircraft (Towbar-less Method).....	9-6
9-4	Aircraft Taxiing	9-8
9-5	Aircraft Taxiing Turning Radii with Minimum Power	9-10
9-6	Propeller/Engine Exhaust Danger Areas.....	9-12
10-1	Engaging the Nose Gear Ground Lock.....	10-2
10-2	Installation of the Main Landing Gear (MLG) Lock Pins.....	10-3
10-3	Static Grounding of the Aircraft.....	10-4
10-4	Mooring the Aircraft at the Nose Landing Gear	10-11

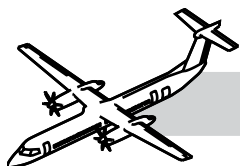
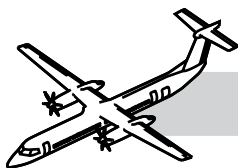


Figure	Title	Page
10-5	Mooring the Aircraft at the Main Landing Gears	10-12
11-1	Left Fuselage Placards and Markings (Example Configuration).....	11-2

TABLES

Table	Title	Page
10-1	Wind Speed and Direction Limitations for Aircraft Mooring	10-10



CHAPTER 5-12

AIRCRAFT GENERAL



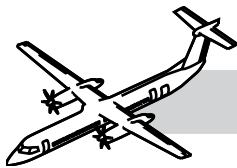
05-00-00 TIME LIMITS/MAINTENANCE CHECKS

The Maintenance Requirements Manual contains the scheduled maintenance tasks performed on the Dash 8 Q400 aircraft. Unscheduled maintenance tasks are included in the Aircraft Maintenance Manual Part 2, Chapter 05.

The Maintenance Requirements Manual (MRM) consists of two parts:

- Part 1 - Maintenance Review Board (MRB) Report
- Part 2 - Airworthiness Limitation Items (ALI)

For non-scheduled maintenance tasks refer to Aircraft Maintenance Manual, Chapter 5.



PART 1 - MAINTENANCE REVIEW BOARD (MRB) REPORT

The MRB report outlines the initial minimum scheduled maintenance/inspection requirements used in the development of an approved maintenance program for the airframe, engines (on-wing), systems and components of the DHC-8 400 aircraft. The MRB requirements are the basis from which each air carrier develops its own maintenance program. The responsible Regulatory Authority inspector shall ascertain that all of the applicable scheduled maintenance/inspection requirements in this report are included in the air carrier's initial continuous airworthiness maintenance program.

REVISION PROCESS

The DHC-8-400 Maintenance Review Board Report is subject to regular review and amendments. Changes may be initiated by, the manufacturer, industry, and/or the MRB. These changes may result from in service experience, manufacturer's test data, analysis and/or changes in utilization or configuration.

MAINTENANCE PROGRAM

The maintenance program lists scheduled maintenance tasks arising from analysis in the following sections:

- **Systems/Powerplant Program:** This section covers all the aircraft systems including the Engine and Propeller. Only Maintenance Significant Items (MSI's) are in this section.
- **Structural Maintenance Program:** This program supplies the air carrier with a basic structural inspection schedule based on Environmental Damage (ED), Accidental Damage (AD), and Fatigue Damage (FD).
- **Zonal Inspection Program:** The Zonal Inspection Program provides a means by which the entire aircraft is examined on a scheduled basis. The program assures that all systems/powerplant, components, installations and structures

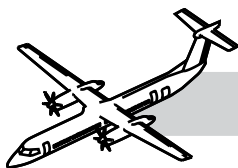
receive adequate inspection to determine security of installation and general condition.

- **Electrical Wiring Interconnection System Inspection Program (EWIS):** Provides a means by which the aircraft wiring is examined on a scheduled basis.
- **Corrosion Prevention and Control Program (CPCP):** A program of maintenance tasks implemented at a threshold designed to control an aircraft structure to Corrosion Level 1 (Corrosion damage that does not require structural reinforcement or replacement) or better.
- **Lightning And High Intensity Radiated Fields (L/Hirf):** This program makes sure that L/HIRF protection is maintained for each Line Replaceable Unit (LRU) and wiring disconnect within a specified system.

Tasks have a specific interval listed as hours, cycles, calendar, engine change, etc. and may have abbreviations as follows:

- Flight Cycles F C
- Flight Hours F H
- Engine Hours E H

An air carrier may do any task listed individually or in packages, provided the interval for each task is not exceeded.



PART 2 - AIRWORTHINESS LIMITATION ITEMS (ALI)

To meet Type certification requirements the certification program identifies maintenance tasks where the task must be performed at or before the required interval. The information contained in Part 2 may change to reflect service and/or test experience or new analysis methods. Any change(s) are generated by Bombardier and approved by Transport Canada.

Five types of maintenance tasks are included in the ALI:

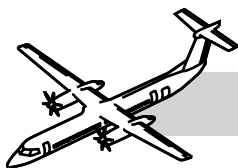
- 1. Certification Maintenance Requirement (CMR):** A CMR is a required task, established during the design certification of the airplane as an operating limitation of the type certificate. CMR's usually result from an analysis conducted to show compliance with catastrophic and hazardous failure conditions. Their intent is to detect significant latent failures that would, in combination with one or more other specific failures or events, result in a hazardous or catastrophic failure condition.
- 2. Structural Maintenance (ALI):** The identification of Fatigue inspection tasks for each Principal Structural Element (PSE) is an integral part of certification for a damage tolerant aircraft. Transport Canada (TC) has a requirement that damage tolerant PSE's cannot go uninspected for more than half of the aircraft design lifetime (40,000 flights).
- 3. Safe Life Component (Discard Tasks):** Operators must monitor the accumulated flight cycles and generate task cards to replace the safe life component, listed in the SAFE LIFE COMPONENTS or SUPPLEMENTS section, at or before a Safe Life Component reaches its Discard Time.
- 4. Fuel System Limitations (FSL):** This section gives a list of the Fuel System Limitations (FSL) that are mandated by the Special Federal Aviation Regulation (SFAR) 88. FSLs are limitation requirements that identify all the necessary maintenance and inspection tasks to

maintain the design features required to preclude the existence or development of an ignition source within the fuel tank system.

- 5. Critical Design Configuration Control Limitations (CDCCL):** Design features that must be preserved to prevent the occurrence of an unsafe condition that was identified in the fuel tank assessment analysis (SFAR 88). The purpose of the CDCCL is to provide instructions to retain the critical ignition source prevention feature during configuration change that may be caused by alterations, repairs or maintenance actions.

UNSCHEDULED MAINTENANCE CHECKS

ATA chapter 05 of the AMM contains unscheduled maintenance tasks that must be completed after non-normal operations have occurred before the aircraft may be returned to service.



06-00-00 DIMENSIONS AND AREAS

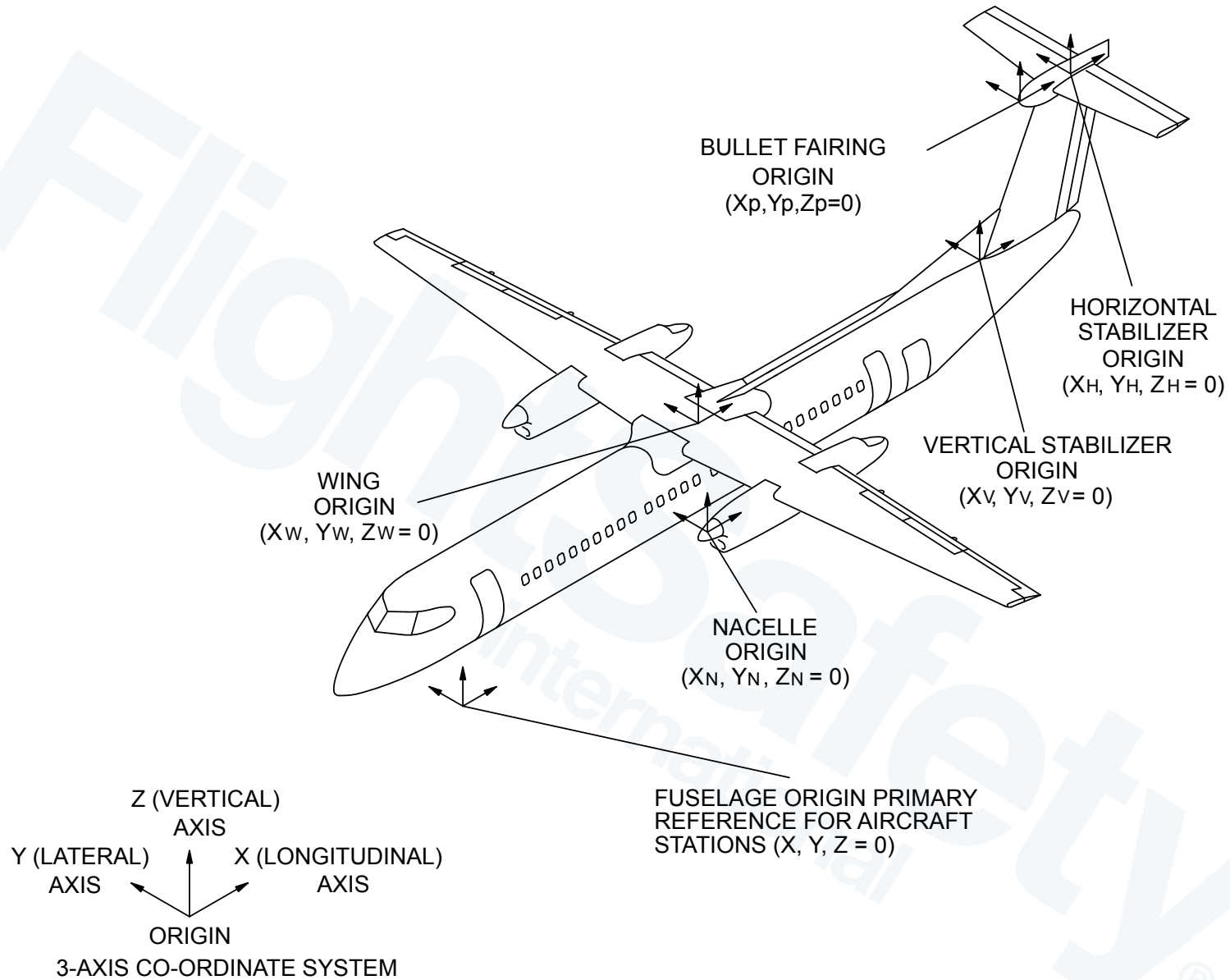
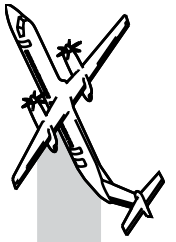
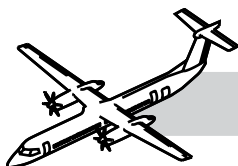


Figure 6-1. Aircraft Axis



06-10-00 DIMENSIONS

GENERAL

Overall Dimensions

Span..... 93 ft - 3 in (28.42 m)
Length..... 107 ft - 9 in (32.83 m)
Height 27 ft - 4 in (8.34 m)
Propeller Ground Clearance... 38.67 in (98.22 m)

06-20-00 AIRCRAFT STATIONS

GENERAL

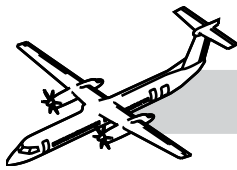
Refer to Figure 6-1. Aircraft Axis.

The three primary axes are given X, Y and Z and the ordinates are identified along these axes from a point of origin. The longitude axis is X, the lateral axis is Y and the vertical axis is Z. The point of origin for the three axes is in front of the forward passenger door, centerline and below the ground reference line of the aircraft. Identification of the ordinates is by the dimension in inches from the point of origin.

The X, Y, Z axis system has more points of origin that are set in the aircraft. These datum points show the location of primary assemblies. The wings, horizontal stabilizer, nacelles and vertical stabilizer are primary assemblies.

These ordinates are identified by:

- XW, YW and ZW (Wings)
- XH, YH, and ZH (Horizontal stabilizer)
- XV, YV, and ZV (Vertical stabilizer)
- XN, YN, and ZN (Nacelles)
- XP, YP and ZP (Bullet fairing).



06-30-00 AIRCRAFT ZONES

GENERAL

Refer to:

- Figure 6-2. Major Zones.
- Figure 6-3. Major Zone 100 (Lower Fuselage) Sub-Zones.
- Figure 6-4. Major Zone 200 (Upper Fuselage) Sub-Zones.
- Figure 6-5. Major Zone 300 (Empennage) Sub-Zone.
- Figure 6-6. Major Zone 400 (Nacelles) Sub-Zones.

The aircraft is divided into eight major zones.

- Zone 100..... Lower fuselage
- Zone 200..... Upper fuselage
- Zone 300..... Empennage
- Zone 400..... Powerplants and nacelles
- Zone 500..... Left wing
- Zone 600..... Right wing
- Zone 700..... Landing gear and landing gear doors
- Zone 800..... Doors

Each major zone is further broken down into subzones for ease of location components and panels. Some examples of these are:

- Subzone 120Flight compartment under floor area
- Subzone 130Fuselage under floor area

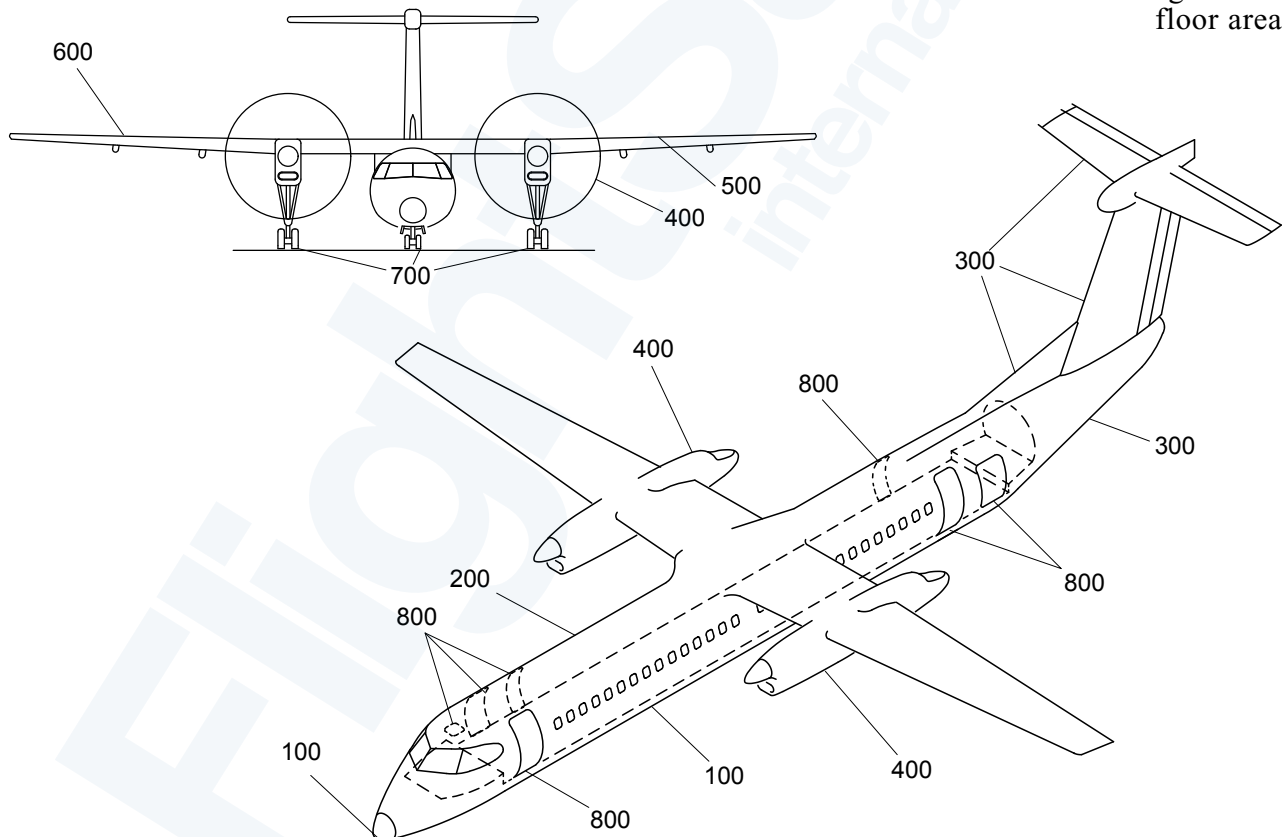


Figure 6-2. Major Zones

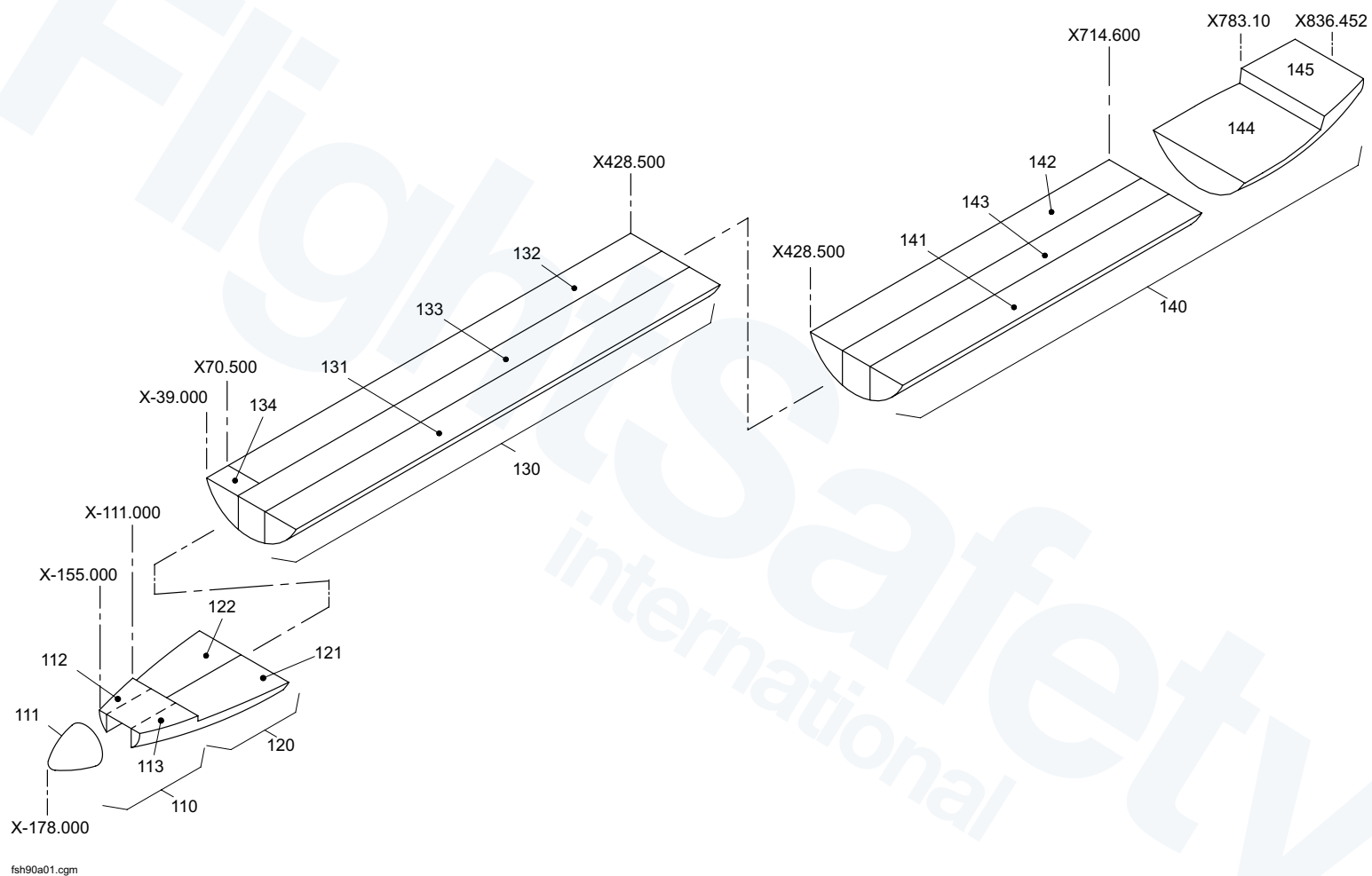
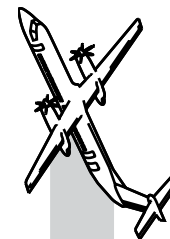


Figure 6-3. Major Zone 100 (Lower Fuselage) Sub-Zones



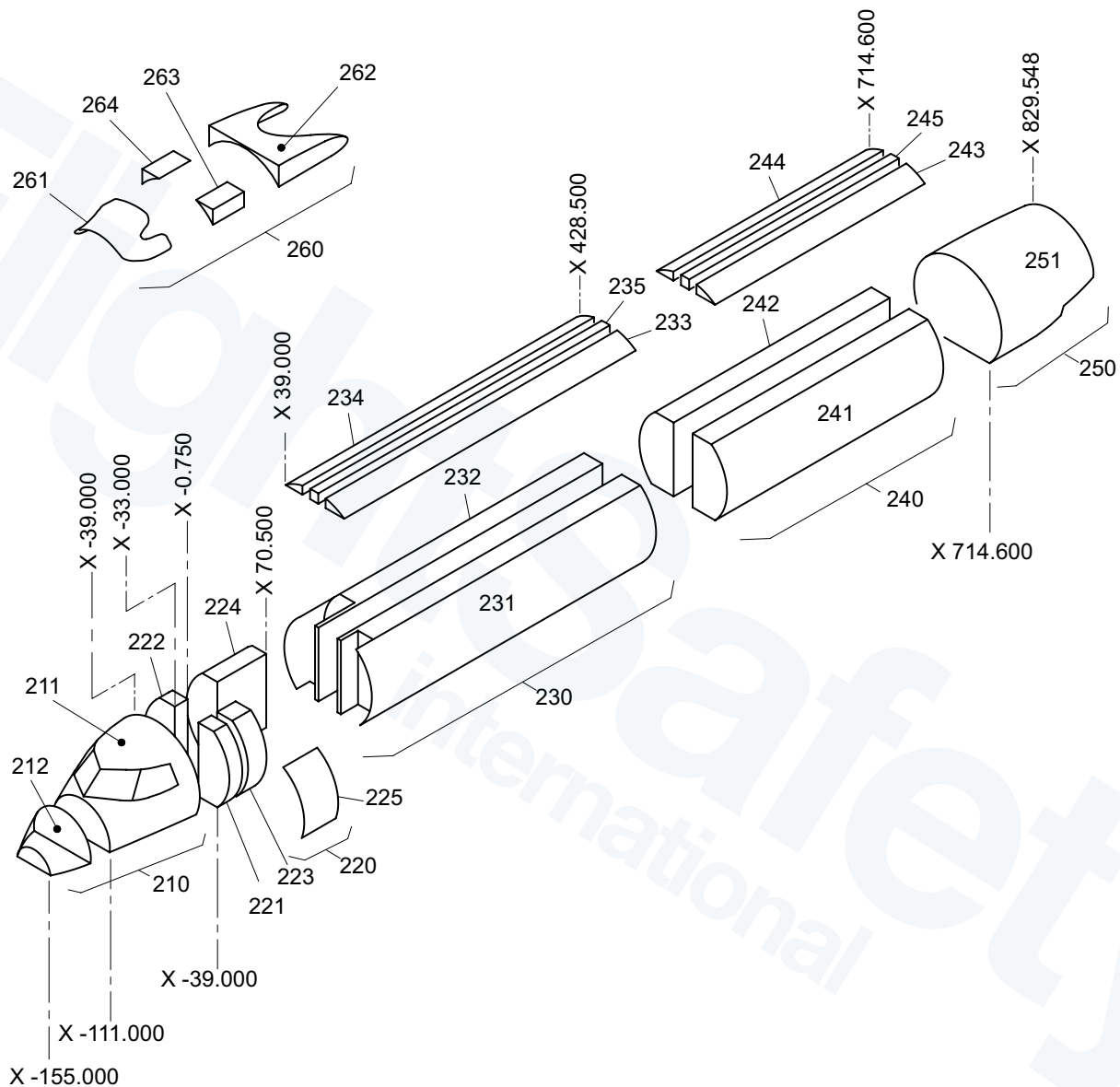
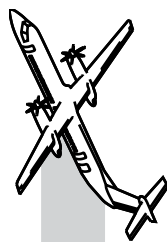


Figure 6-4. Major Zone 200 (Upper Fuselage) Sub-Zones

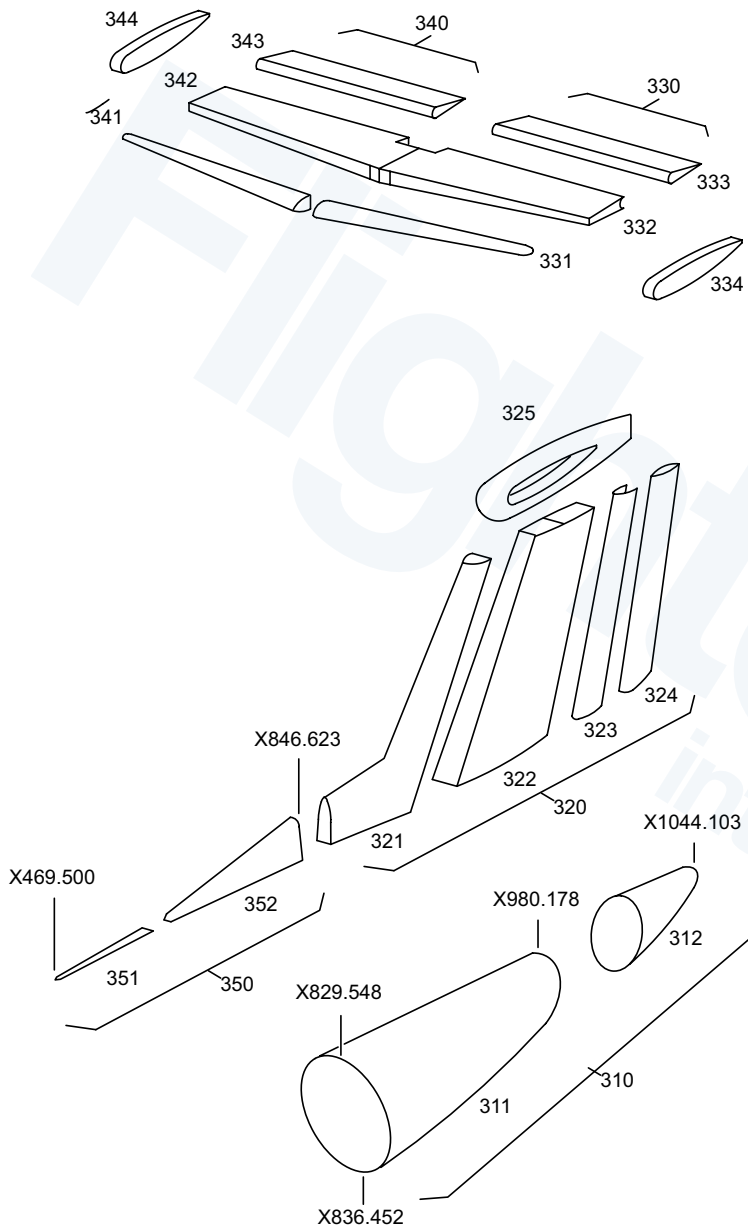


Figure 6-5. Major Zone 300 (Empennage) Sub-Zone

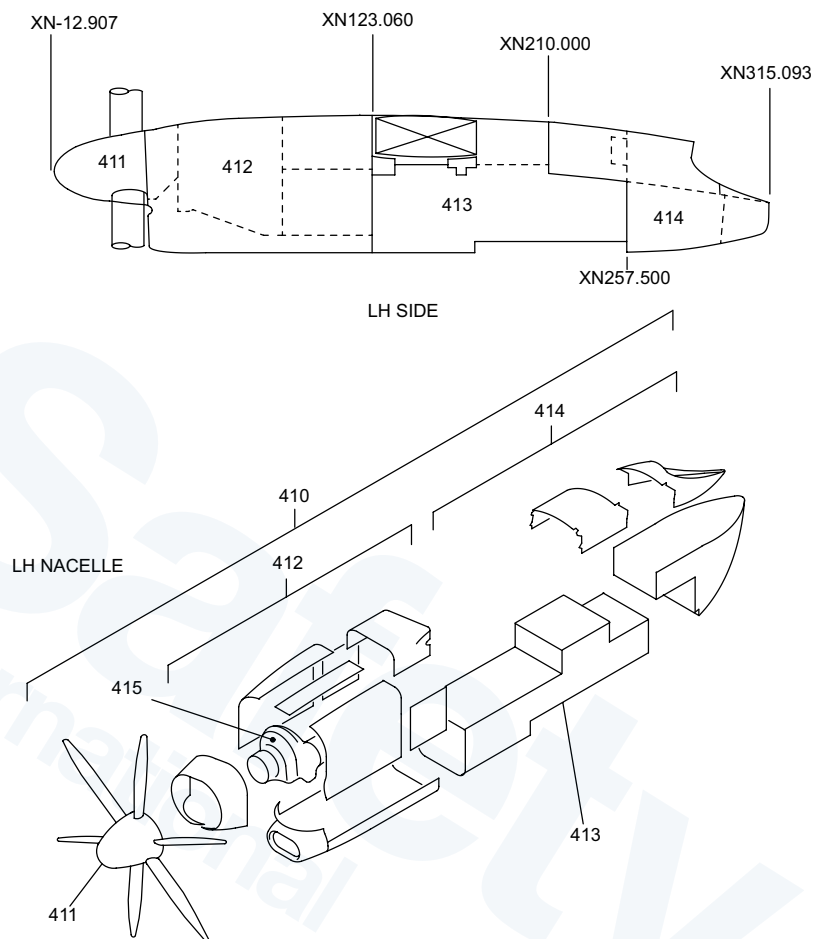
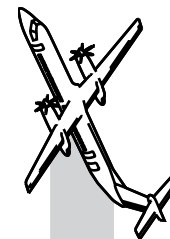
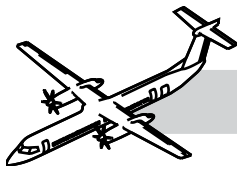


Figure 6-6. Major Zone 400 (Nacelles) Sub-Zones





06-41-00/06-43-00 EXTERNAL & INTERNAL ACCESS PANELS AND DOORS

GENERAL

Refer to Figure 6-7. Access Panels and Doors.

The access panels and doors are identified by an alphanumeric code. The identification code makes it easier to find the different access panels, doors and fairings. Each panel, door and fairing has a two-group code to indicate its position on the aircraft.

A three digit numeric group indicates the Major zone and sub zone where the panel, door or fairing is located. A two digit letter group gives the position of the panel, door or fairing in relation to the aircraft as follows:

- The first letter identifies the general position. This starts with the letter A and goes from the inboard to outboard and from the front to the rear.
- The second letter identifies the location of the panel, door or fairing in relation to the aircraft as follows:
 - T (Top)
 - B (Bottom)
 - L (Left side)
 - R (Right side)
 - Z (Internal).

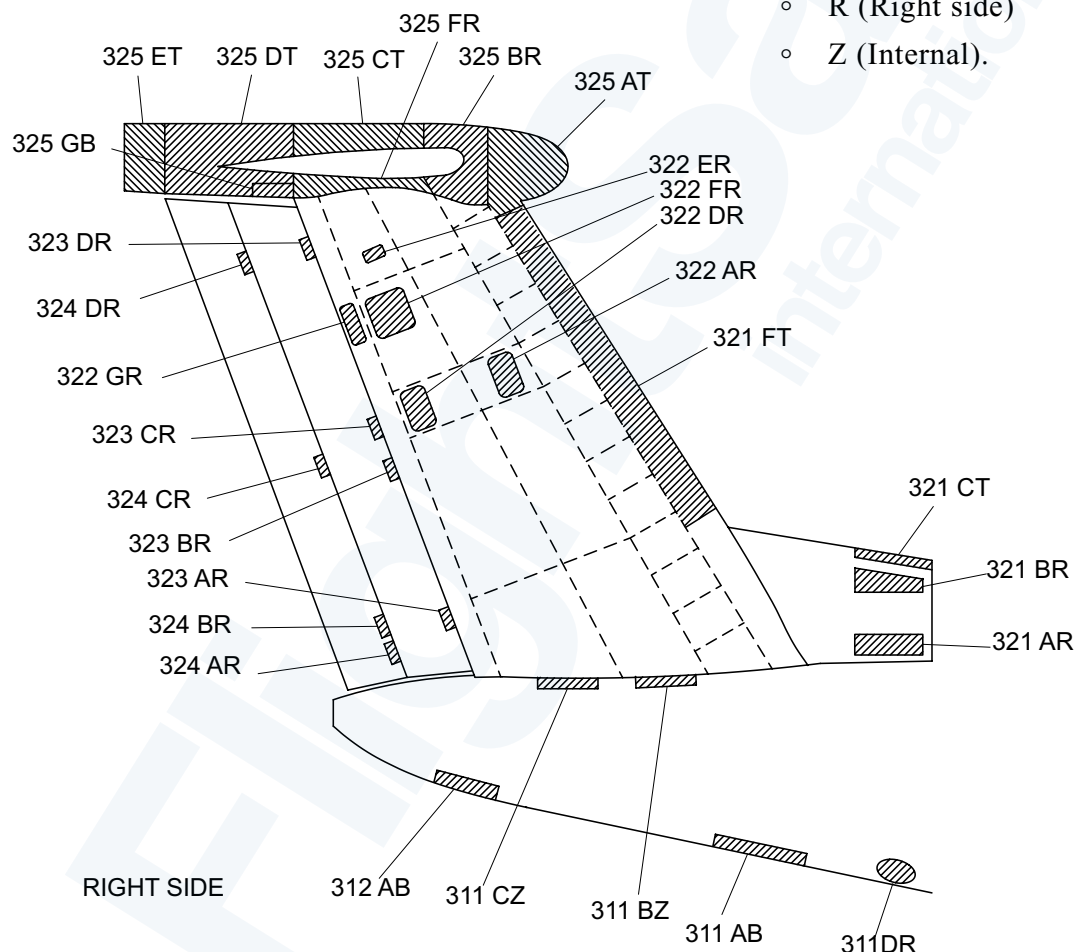
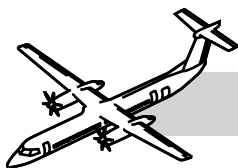
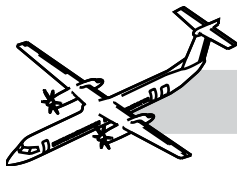


Figure 6-7. Access Panels and Doors



07-00-00 LIFTING AND SHORING



07-00-00 LIFTING AND SHORING

INTRODUCTION

There are three jacking points on the aircraft structure. There are two jacking points on the wing rear spar and one jacking point on the forward fuselage, aft of the nose landing gear (NLG).

There is a jacking point at the nose landing gear axle and another at each of the two main landing gear axles.

GENERAL

Refer to Figure 7-1. Jacking of the Aircraft.

The complete aircraft can be lifted without payload (engines and propellers or equivalent weight installed).

The engines can be removed when the aircraft is on the jacks. Ballast is distributed equally on the floor of the cockpit / cabin area forward of the lavatory (see MTM Chapter 71).

If necessary, the aircraft nose landing gear (NLG) can be lifted only using the structural jacking point aft of the NLG. The two main landing gears (MLG) must stay on the ground. The aircraft weight must be less than 48,792 lb (22,132 kg) and the areas above and below the aircraft are clear.

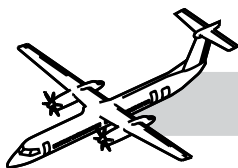
The aircraft can be lifted with axle jacks for the nose gear or for the two main gears.

The aircraft can be lifted at the nose fuselage and the main landing gear Axles for weighing the aircraft.

If one of the main landing gear OLEO is depleted of its nitrogen and oil and requires replacement of the lower bearing seal, the wing of the effected main landing gear can be lifted until both wings level is achieved.

To lift the aircraft on all three jacks at the wing and forward fuselage, the procedure is as follows:

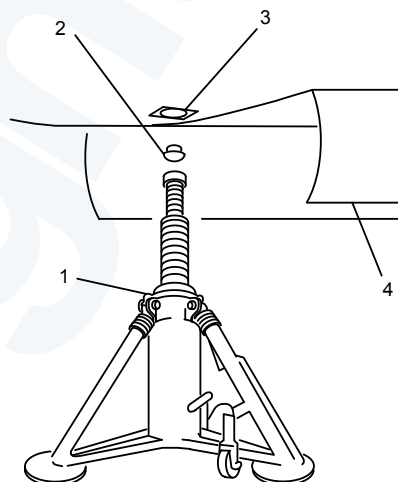
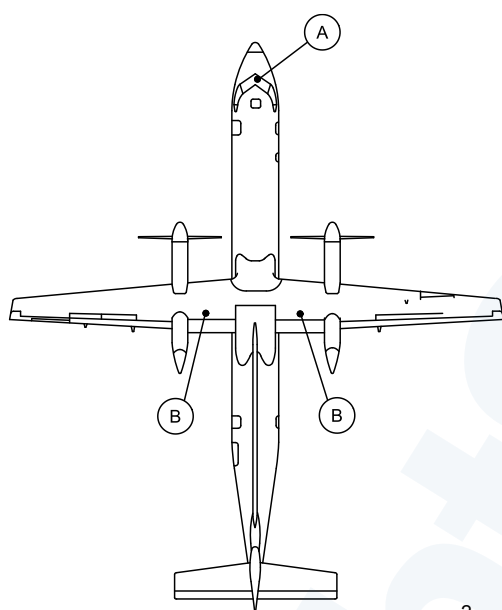
- The top door of the forward emergency exit is removed and the lower door is opened
- A plumb bob line is put through the hole in the plumb bob hanger
- The parking brake is released
- The chocks are moved clear of the nose landing gear (NLG) tires
- A nose fuselage jack is put at the nose fuselage jacking point
- The front fuselage jack adapter is put in the front fuselage receptacle and held there
- The jack height is increased until the jack correctly engages the adapter
- A tail stand is positioned approximately 4 to 6 in. (101 to 152 mm) from the adapter on the bottom of the AFT fuselage
- The nose jack is raised continuously and the NLG jack locking collar is moved down while the aircraft is lifted
- The plumb bob is monitored and lifting of the aircraft is stopped when the plumb bob aligns with the longitudinal level mark on the target plate
- A wing jack is put at each wing jacking point
- The wing jack adapter is put in the right wing receptacle and held there and the jack height is increased until the jack touches the adapter.
- The wing jack adapter is put in the left wing receptacle and held and the jack height is increased until the jack touches the adapter
- The aircraft is carefully lifted on the jacks at the three positions, at the same time
- The jacks are adjusted as required to align with the plumb bob



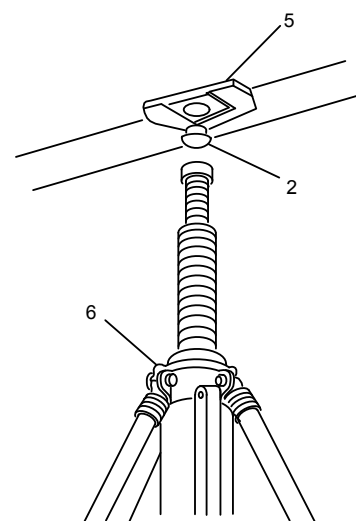
- The NLG jack locking collar is tightened and wing jack locking collar on each jack to lock the jack extensions
- A tail stand is attached to the bottom of the aft fuselage and the tail stand extension is locked.

The aircraft is lowered as follows:

- The tail stand is removed from the bottom of the aft fuselage
- The NLG jack locking collar is loosened and the wing jack locking collar on each jack to unlock the jack extension.
- The three jacks are lowered gradually and at the same time until the main landing gear has weight on wheels
- The wing jack adapters are removed from the wing receptacles
- The wing jacks are moved clear of the aircraft
- The nose fuselage jack is lowered until the nose landing gear has weight on wheels
- The front fuselage jack adapter is removed from the receptacle of the NLG jacking point
- The nose fuselage jack is moved clear of the aircraft
- Chocks are put at the NLG and MLG tires
- The parking brake is set.



(A) NOSE FUSELAGE JACKING POINT

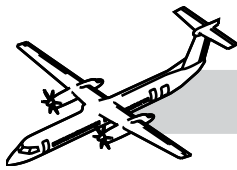


(B) WING JACKING POINTS

LEGEND

1. NLG Jack Locking Collar
2. Jack Adapter
3. Nose Fuselage Jack Point
4. NLG Door (Reference)
5. Wing Jack Point
6. Wing Jack Locking Collar

Figure 7-1. Jacking of the Aircraft



Refer to Figure 7-2. Jacking of the Aircraft at the NLG Axle.

A jack at the NLG axle is used to lift the aircraft as follows:

- The nosewheel centerd and caster lock-pin installed
- The nosewheel steering is set off
- The NLG axle jack is put at the NLG jacking point
- The jack height is increased until the jack correctly engages the jacking point
- The axle jack is used to lift the aircraft until the related wheel is approximately 2 in. (5.1 cm) clear of the ground.

The aircraft is lowered as follows:

- Lower the aircraft until the NLG has weight on wheels
- The jack is moved clear of the aircraft.
- Caster lock-pin removed.

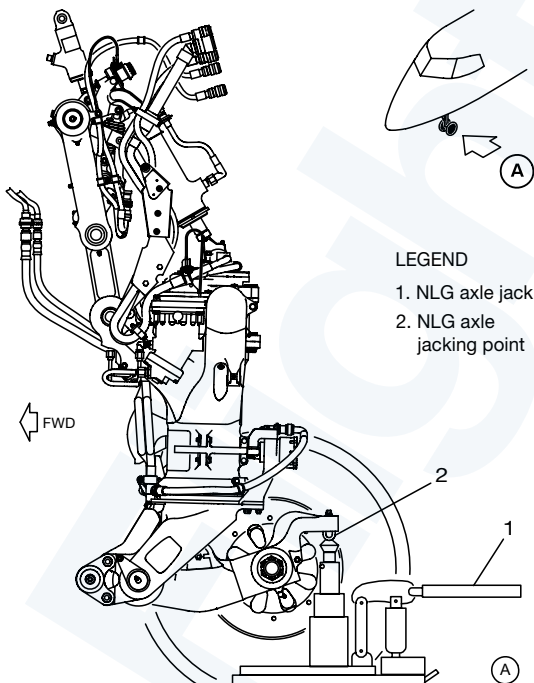


Figure 7-2. Jacking of the Aircraft at the NLG Axle

Refer to Figure 7-3. Jacking of the Aircraft at the MLG Axle.

A jack at the MLG axle is used to lift the aircraft as follows:

- The MLG axle jack is put at the MLG axle jacking point
- The jack height is increased until the jack correctly engages the jacking point
- The axle jack is used to lift the aircraft until the related wheels are approximately 2 in. (5.1 cm) clear of the ground.

The aircraft is lowered as follows:

- Lower the aircraft until the MLG has weight on wheels
- Move the jack clear of the aircraft.

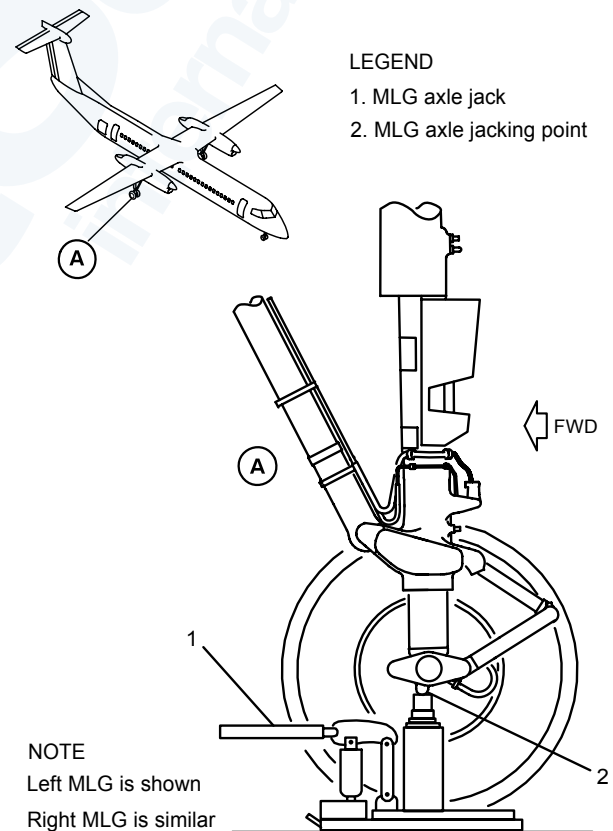
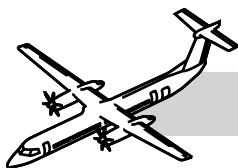


Figure 7-3. Jacking of the Aircraft at the MLG Axle



07-00-00 SPECIAL TOOL & TEST EQUIPMENT

- GSB0700007 Front Fuselage Adapter Jack (Qty 1)
- GSB0700016 NLG Axle Jack
- GSB0700017 Wing Adapters Jack (Qty 2)
- GSB0700019 Wing Jacks (Qty 2)
- GSB0700020 Front Fuselage Jack
- GSB0700023 MLG Axle Jack
- GSB0700024 Tail-stand
- GSB1000031 Wheel Chocks MLG & NLG
- GSB0800001 Plumb Bob Aircraft Leveling
- GSB0700023 MLG axle jack.

07-00-00 CAUTIONS & WARNINGS

CAUTION

If it is possible, defuel the aircraft to decrease the weight. This will help to prevent damage to the aircraft structure.

CAUTION

Do not lift the aircraft in an open area if the wind speed is more than 20 mph (32 km/h).

WARNING

Do not lift the aircraft on jacks if you think the load at the nose-fuselage jacking point will be 4700 lbs (2132 kg) or more.

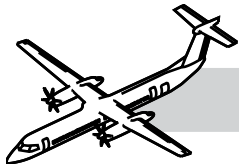
WARNING

Do not lift the aircraft at the jacking points of the wing and the nose fuselage if the weight is more than 48,792 lbs (22,132 kg).

WARNING

Before you lift the aircraft on jacks, make sure that:

- No person is in/on the aircraft
- You release the parking brake
- You remove the chocks from the wheels
- There is sufficient clearance around/above the aircraft.

**WARNING**

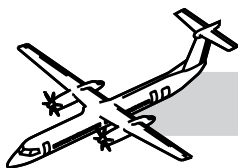
Before you lower the aircraft on jacks, make sure that:

- No person is in/on the aircraft
- You release the parking brake
- You engage the down lock of the nose landing gear
- You install the ground lock-pins of the main landing gear
- There is sufficient clearance around/above the aircraft.

If you do not do this, you can cause injuries to persons and damage to the equipment.

WARNING

Make sure you remove the jack adapter from the wing immediately after the jack is clear of the adapter.



08-00-00 LEVELING AND WEIGHING

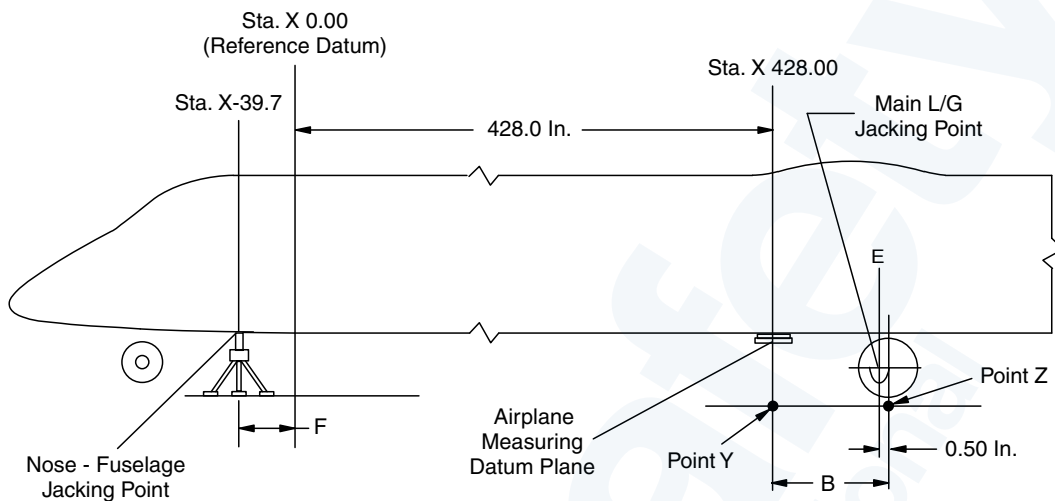
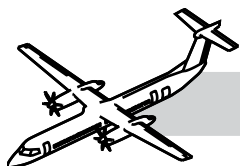


Figure 8-1. Weighing of the Aircraft

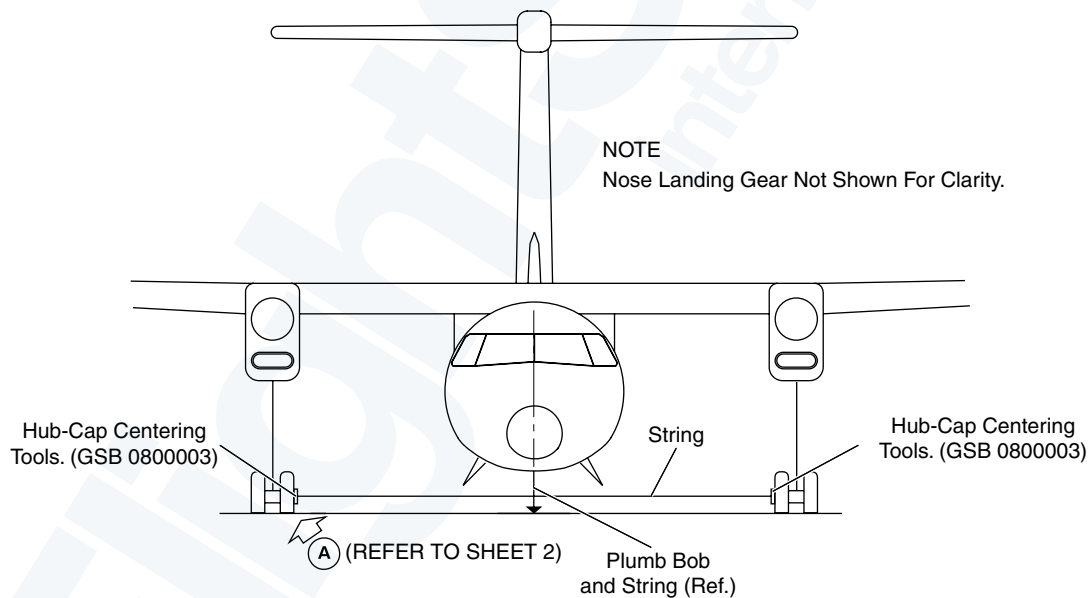
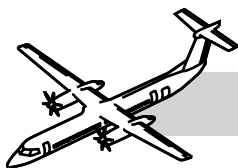


Figure 8-2. Method to Determine the Center Line of Main Wheels



08-10-00 WEIGHING OF THE AIRCRAFT

INTRODUCTION

Electronic load sensing devices, at the nose fuselage and the axles of the main landing gear, must be used to weigh the aircraft. The weights are recorded to calculate the total weight and the center of gravity of the aircraft.

The basic weight of the aircraft is that weight that includes all fixed operating equipment (both standard and optional), trapped and unusable fuel, and full engine oil.

To calculate the weight and the center of gravity of the aircraft, refer to PSM 1-84-8, Weight and Balance Manual.

If it is necessary to use the metric system, the same procedure is applicable. To calculate the metric weight and the center of gravity of the aircraft, refer to PSM 1-84-8M, Weight and Balance Manual (Metric).

GENERAL

Refer to:

- Figure 8-1. Weighing of the Aircraft.
- Figure 8-2. Method to Determine the Center Line of Main Wheels.

Before the aircraft is weighed it must be at the basic weight and in the configuration as follows:

- The landing gear is extended
- The MLG lockpins are installed and the nose-landing-gear ground lock is engaged
- The flaps are fully retracted
- The flight controls are in the neutral position with the gust lock applied

- All doors and panels are installed and closed (the forward right emergency exit door not included)
- The aircraft is at a level position
- Make sure the external, internal areas and the windows are clean and dry
- The food supplies are removed from the galley
- The galley liquid containers are drained
- All emergency equipment is stowed in its correct location.
- The aircraft is pressure defueled
- The fuel left in the tanks and surge bays are drained through their related drain valves on the bottom of the wings
- Make sure that the engine oil is full
- Make sure that the reservoirs for the No. 1 and No. 2 hydraulic systems are full
- Make sure that the reservoir for the No. 3 hydraulic system is full
- Make sure that the reservoir for the auxiliary hydraulic is full
- The lavatory is serviced as necessary
- The wash water is serviced as necessary.

Ensure that all tools and equipment not included in the basic weight configuration are removed (Refer to PSM 1-84-8, Weight and Balance Manual, Chapter 2).

Weigh the aircraft on jacks with the electronic load sensing Kit as per PSM 1-84-8, Weight and Balance Manual, Chapter 2.

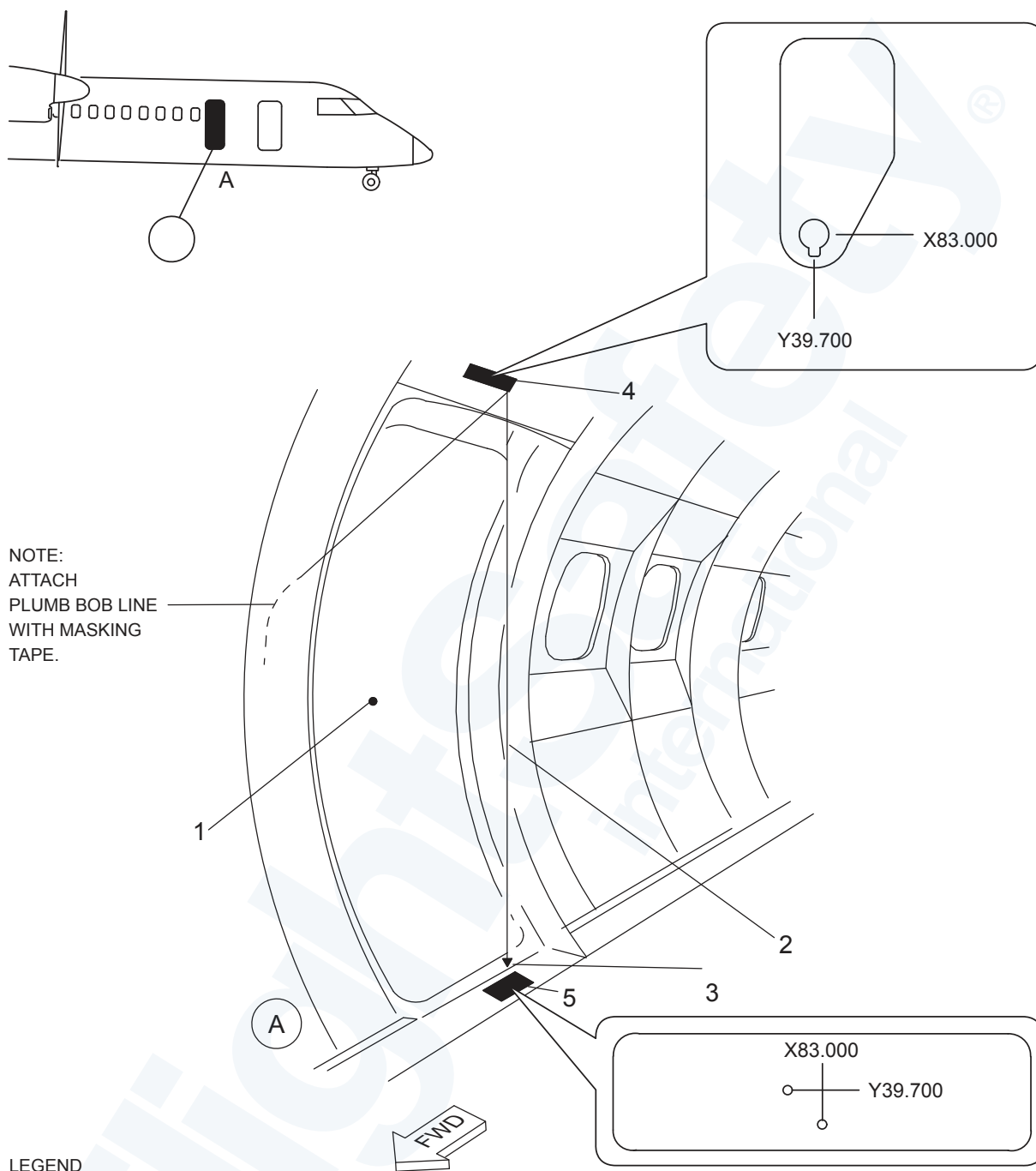
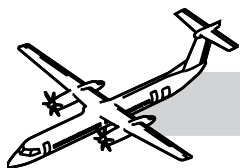
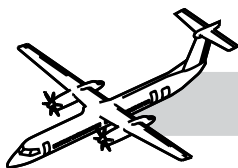


Figure 8-3. Leveling of the Aircraft



08-20-00 LEVELING OF THE AIRCRAFT

NOTES

INTRODUCTION

A plumb bob hanger and a target plate are used to make the aircraft level.

The plumb bob hanger is on the top surface of the Type II/III emergency-exit-door on the right side of the fuselage. The target plate is on the bottom surface of the emergency exit door. The center mark of the target plate is at STA. X83.00 and Y39.70.

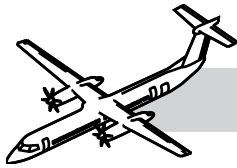
The aircraft level must be made by adjustment of the jack heights. The jacks are put at the nose fuselage and the axles of the main landing gear.

GENERAL

Refer to Figure 8-3. Leveling of the Aircraft.

The aircraft is leveled as follows:

- The upper door of the forward right emergency exit is removed
- The lower door is opened
- The filler panel at the top of the opening of the emergency exit door is removed to get access to the plumb bob hanger
- The plumb bob line is put through the hole in the hanger
- Masking tape is used to attach the plumb bob line to the external fuselage at the forward side of the emergency exit door.
- The point of the plumb bob is put 0.25 in (0.64 cm) or less above the target plate
- Jacks are put at the jacking points of the nose fuselage and the axles of the main landing gear
- The jacks are adjusted until the plumb bob aligns with the lateral mark on the target plate.



08-00-00 SPECIAL TOOL & TEST EQUIPMENT

- GSB0811001 Electronic Weighing kit
- GSB0800003 Main Wheel Measuring Adapters
- GSB1210015 Fuel Sample Unit - 1/4 Quart
- GSB2810001 Fuel Drain Valve Tool
- GSB0700024 Tailstand
- GSB0800001 Plumb Bob Aircraft Leveling.

08-00-00 CAUTIONS & WARNINGS

WARNING

Do the leveling and weighing procedure in a closed hangar. If you do not do this, wind can cause incorrect results.

WARNING

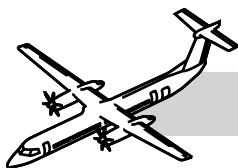
Do not use the wing jacking points to lift the aircraft when you do the leveling and weighing procedure. This will cause side loads to occur, which will give incorrect indications from the load-cells.

WARNING

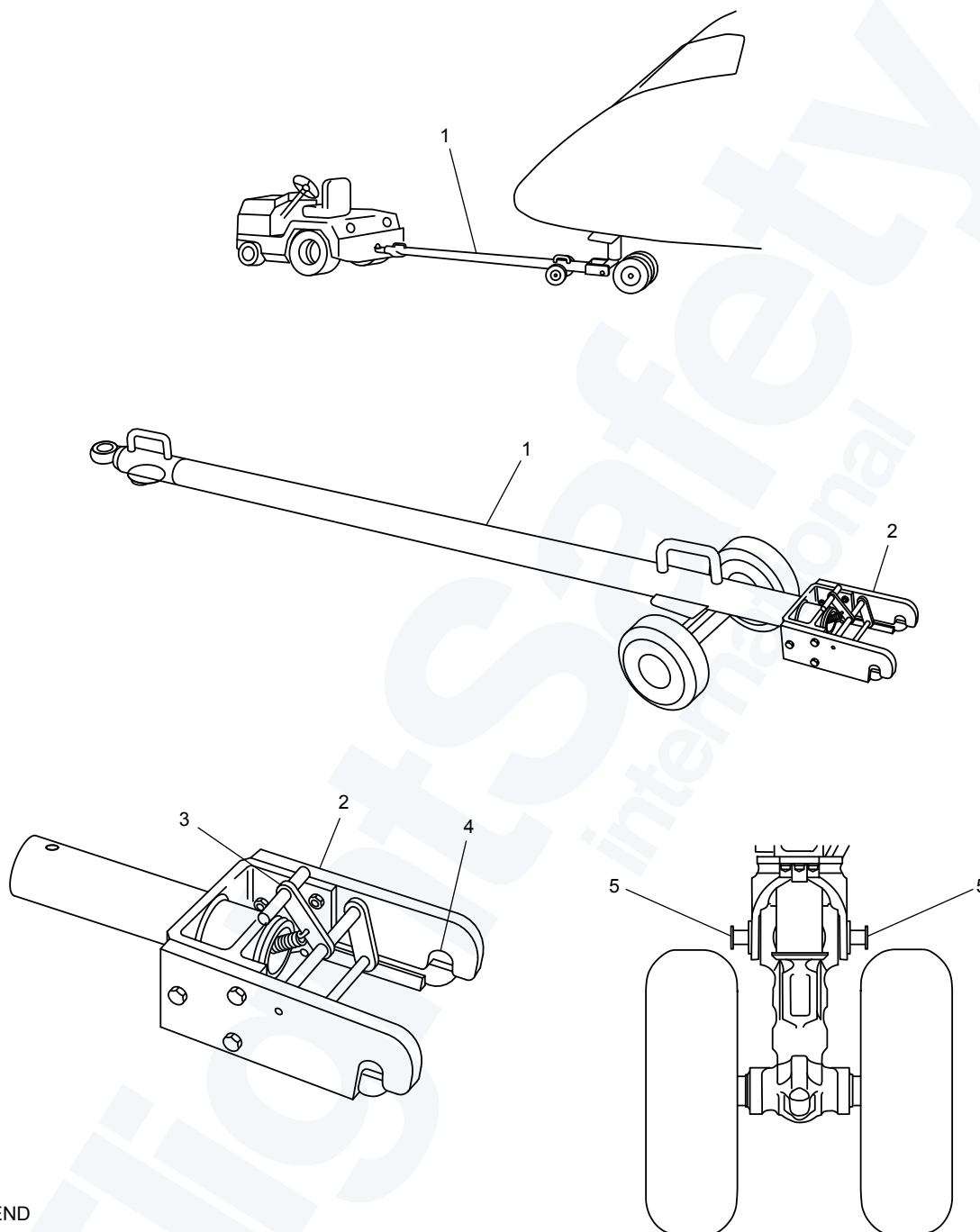
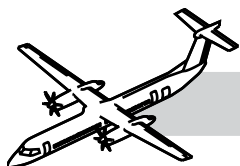
Failure to follow all procedures and safety procedures may result in the aircraft not being in a safe balanced condition for take-off flight and landing.

WARNING

Do not use the wing jacking points to lift the aircraft when leveling and weighing. If you do, you will cause side loads to occur. This will give you incorrect indications from the load-cells and the aircraft may not be in a safe balanced condition for take off, flight, and landing.



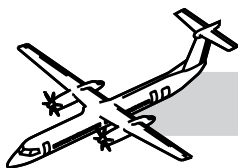
09-00-00 TOWING AND TAXIING



LEGEND

1. Tow Bar.
2. Towhead.
3. Locking Handle.
4. Tow Spool Attachment Point.
5. Towing Spool.

Figure 9-1. Towing the Aircraft with a Towbar



09-10-00 TOWING OF THE AIRCRAFT

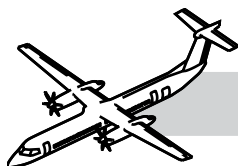
INTRODUCTION

The aircraft is towed or pushed from the nose landing gear with a tow vehicle and a towbar or with a towbar-less tow vehicle.

GENERAL

Refer to Figure 9-1. Towing the Aircraft with a Towbar.

- A pressure check of the parking brake system is done at the accumulator gauge in the right wing root. Make sure that the pressure is a minimum of 1000 psi (6895 kPa).
- The engine lower cowls for both the engines are installed and secured.
- If towing is to be done with one or both engines removed, make sure that ballast weights are loaded and uniformly distributed on the floor of the cockpit and forward cabin area forward of the lavatory.
- The grounding wire from the aircraft is disconnected.
- The external servicing connections are removed from the aircraft.
- A visual inspection of the landing gear tires is done.
- All external doors of the aircraft are closed
- The Nosewheel Steering is set to OFF.
- The towbar head is connected to the towing spools. Connect the towbar eye to the tow vehicle.
- The chocks are removed from the landing gear wheels.
- The Parking Brake is set to OFF.
- The aircraft is slowly towed from its parked position.
- Make sure the aircraft is not towed at more than a maximum speed of 5 mph (8.05 kph).
- The aircraft turning radii is kept in the limits of 120 °s each side of center (Painted arrows on the two aft NLG doors show these limits).
- The parking brake is engaged after the aircraft is towed.
- The nosewheel is straight before the towbar is disconnected.
- The towbar eye is disconnected from the tow vehicle.
- The towbar head is disconnected from the towing spools.
- The aircraft is parked.



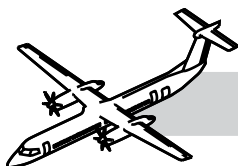
ITEM NO.	MAIN GEAR CONDITION	TIRE FOOTPRINT EXAMPLES	IS TAXIING ALLOWED?	IS TOWING ALLOWED?	DISTANCE ALLOWED (TAXIING & TOWING)	NOSE WHEEL ANGLE OF TURN	REMARKS
1	ONLY ONE FLAT TIRE (ANY TIRE)		YES	YES	UNLIMITED	NORMAL	SEE NOTES 1., 2., 6., 8.
2	TWO FLAT TIRES (ONE ON EACH AXLE)		YES	YES	UNLIMITED	NORMAL	SEE NOTES 1., 3., 4., 6., 8.
3	TWO FLAT TIRES (ON ONE AXLE)		YES	YES	MINIMUM TO CLEAR RUNWAY	10° MAXIMUM	SEE NOTES 1., 3., 4., 5., 6., 7., 9.
4	THREE FLAT TIRES (ANY COMBINATION)		YES	YES	MINIMUM TO CLEAR RUNWAY	10° MAXIMUM	SEE NOTES 1., 3., 4., 5., 6., 7., 9.
5	FOUR FLAT TIRES		YES	YES	MINIMUM TO CLEAR RUNWAY	10° MAXIMUM	SEE NOTES 1., 3., 4., 5., 6., 7., 9.
	NOSE GEAR CONDITION						
6	ONE FLAT TIRE		YES	YES	UNLIMITED	NORMAL	SEE NOTES 1., 2., 4., 8.
7	TWO FLAT TIRES		YES	YES	MINIMUM TO CLEAR RUNWAY	10° MAXIMUM	SEE NOTES 1., 3., 4., 5., 6., 7., 8. or 9.

= FLAT TIRE

NOTES

1. Avoid sharp turns, abrupt starts and stops.
2. Maximum taxiing or towing speed = 5 mph (8 kmh).
3. Maximum taxiing or towing speed = 2 mph (3 kmh).
4. After clearing runway, or if additional tire fails, aircraft should be stopped and serviceable wheel/tire assembly(ies) installed to satisfy conditions in Item No. 2 or 6.
5. Taxiing or towing with two flat tires on same gear can result in wheel damage.
6. After any tire failure or excessive heat condition, the affected wheel assembly must be inspected per applicable Vendor Manual prior to further use.
7. Under multiple failed tire condition, the affected landing gear assemblies and linkages must be inspected for possible structural damage.
8. Towbar required.
9. Use of rope on both main landing gear required.

Figure 9-2. Towing/Moving Aircraft with Flat Tires



Refer to Figure 9-2. Towing/Moving Aircraft with Flat Tires.

NOTES

An aircraft with a flat tire or a combination of flat tires, on the runway, taxiway or apron, is allowed to be moved with the use of towbar or using suitable rope to clear the runway or taxiway using the following procedure:

- Make sure that the rims of the wheel with flat tires are not damaged and the aircraft is still supported by its own main and nose landing gears
- Make sure the nose landing gear ground lock is engaged
- Make sure the main landing gear ground lock pins are installed
- The precautions that follow are obeyed when an aircraft is towed with two flat tires on one or more axles:

Refer to the table that follows for maximum allowable towing loads with flat tires:

Maximum Allowable Towing Load for ONE deflated tire (lb)	Maximum Allowable Towing Load for TWO deflated tires (lb)
7185	5987

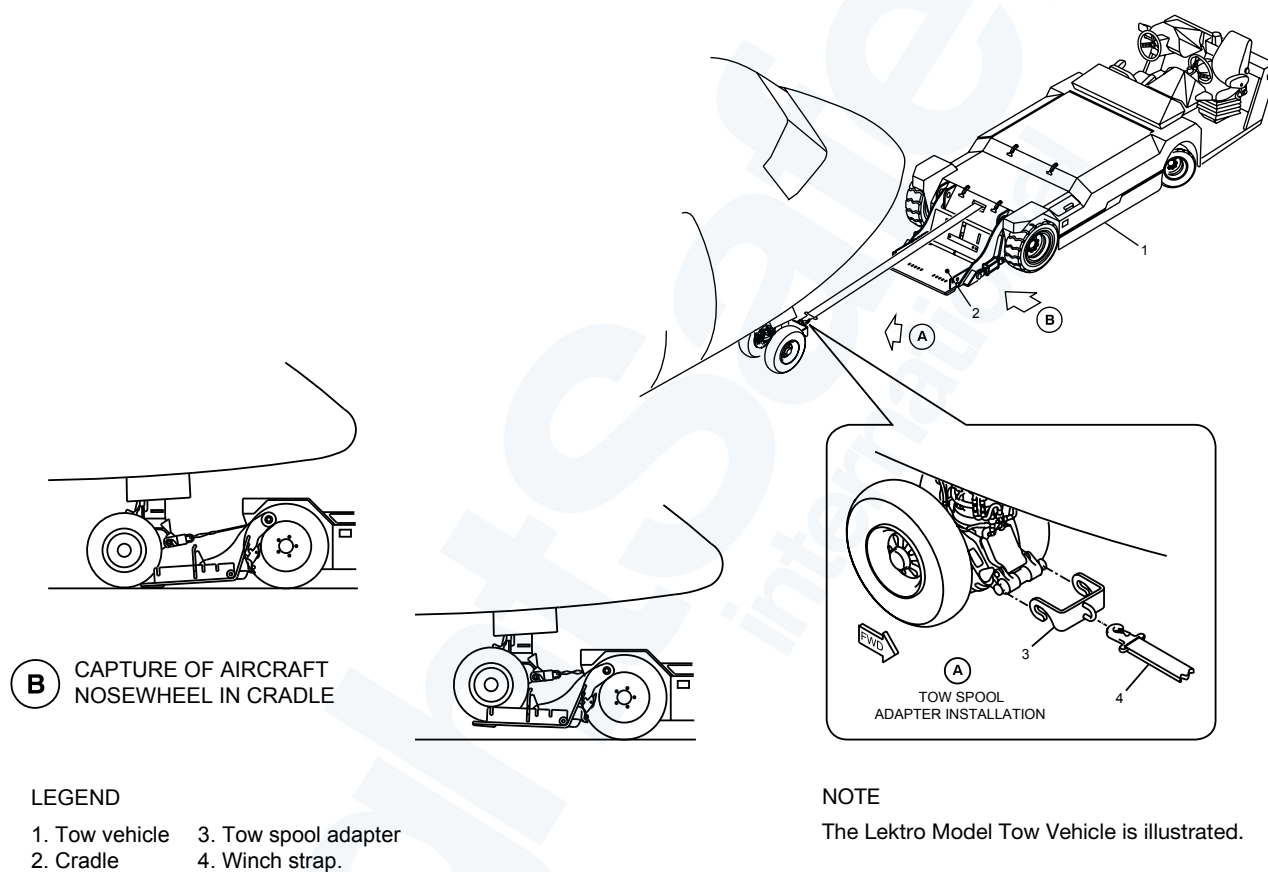
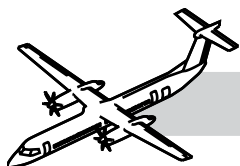
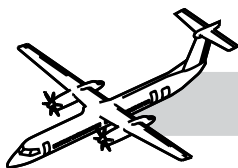


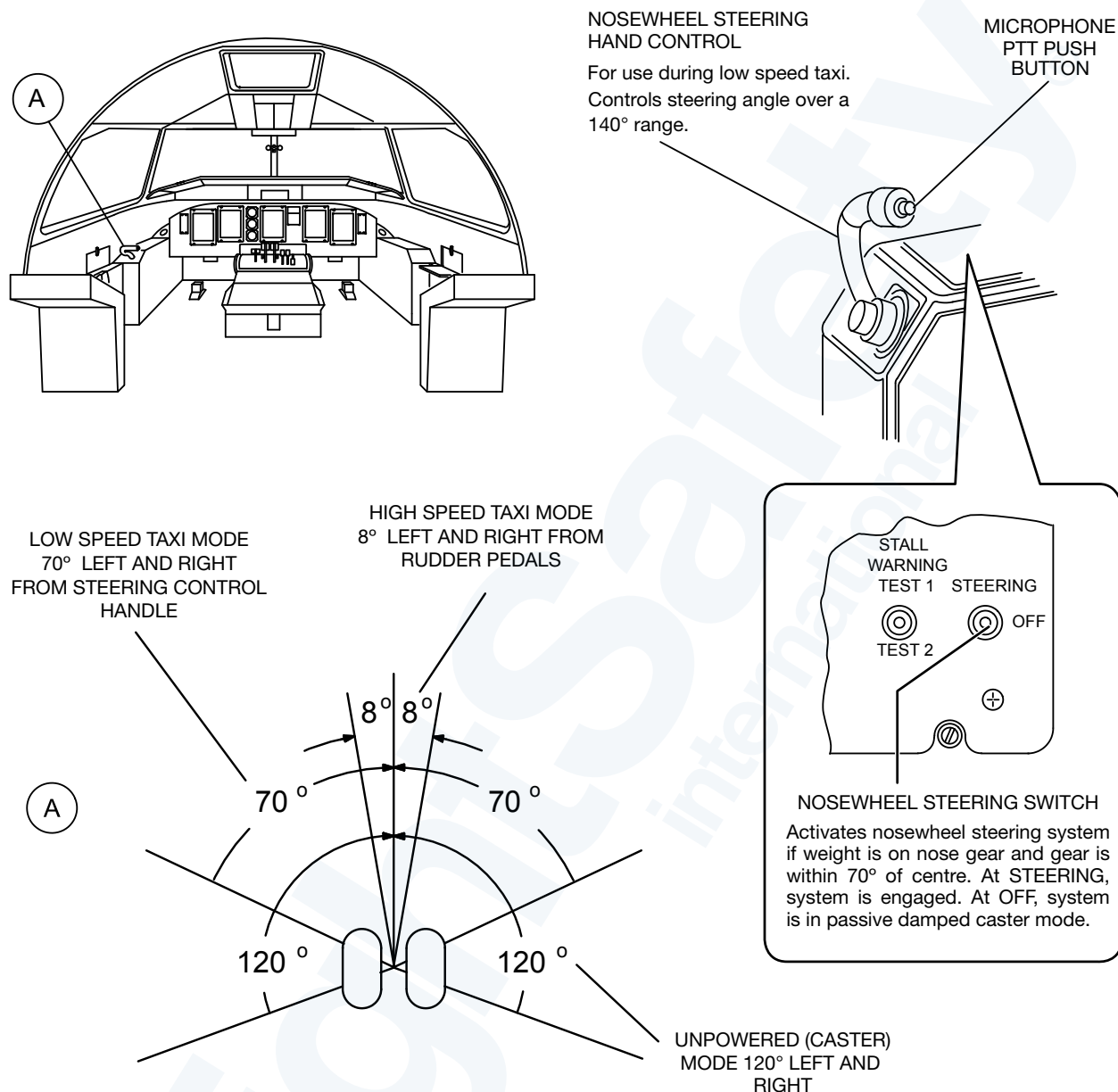
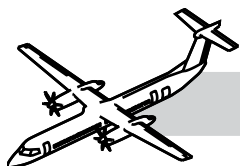
Figure 9-3. Towing of the Aircraft (Towbar-less Method)



Refer to Figure 9-3. Towing of the Aircraft (Towbar-less Method).

The aircraft is towed as follows:

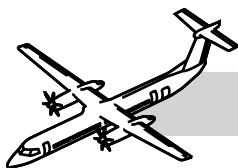
- A pressure check of the parking brake system is done at the accumulator gauge in the right wing root. Make sure that the pressure is a minimum of 1000 psi (6895 kPa).
- The engine lower cowls for both the engines are installed and secured.
- If towing is to be done with one or both engines removed, make sure that ballast weights are loaded and uniformly distributed on the floor of the cockpit and forward cabin area forward of the lavatory.
- The grounding wire from the aircraft is disconnected
- The external servicing connections are removed from the aircraft
- A visual inspection of the landing gear tires is done
- All external doors of the aircraft are closed
- The Nosewheel Steering is set to OFF
- The large rear gate is installed in the applicable slot on the nosewheel cradle of the tow vehicle
- The vehicle is moved to the front of the aircraft such that the nosewheel cradle is aligned with the nosewheel
- The tow spool adapter is installed on the NLG tow spools
- Make sure that the tow vehicle is parked with the brakes on
- The winch strap is connected to the adapter
- The parking brake on the aircraft is set to OFF
- The aircraft is winched on the cradle
- The cradle is lifted sufficiently to prevent damage from obstacles while the aircraft is towed
- The aircraft is slowly towed from its parked position
- The aircraft turning radii is kept in the limits
- Large turns are made when possible
- Changes to speed or directions are done slowly
- Turn limits must not be more than 90° right and left from the center
- The parking brake is engaged, as necessary, after the aircraft is towed
- The nosewheel is set straight
- The cradle is lowered to the ground level and release the tension on the winch strap
- The tow spool adapter is removed from the NLG tow spools
- The tow vehicle is slowly moved away from the aircraft.



THE NOSEWHEEL STEERING SYSTEM INCORPORATES THREE MODES OF OPERATION FOR TAXIING:

1. A high speed mode, controlled by the rudder pedals, gives a steering angle of 8° each side of centre.
2. A low speed mode, controlled by a steering hand control found on the pilot's side console, gives a steering angle of 70° each side of centre.
3. A free casting (unpowered) mode, controlled by differential engine thrust and/or braking gives a steering angle of 120° each side of centre.

Figure 9-4. Aircraft Taxiing



09-20-00 TAXIING OF THE AIRCRAFT

INTRODUCTION

The aircraft is moved with engine power, Nosewheel steering, and brakes by a crew member in the pilot seat.

GENERAL

Refer to:

- Figure 9-4. Aircraft Taxiing.
- Figure 9-5. Aircraft Taxiing Turning Radii with Minimum Power.

The aircraft is taxied as follows:

- Make sure that the tires are serviceable
- The pressure of the parking brake system on the accumulator gauge installed in the right wing root is checked
- Make sure that the pressure is a minimum of 1000 psi (6895 kPa).
- The parking brake is set
- The ground wire is removed from the aircraft
- The chocks are removed from the landing gear
- All external doors of the aircraft are closed
- Make sure that the area around the aircraft is clear of personnel and contamination that may get ingested in the engines
- The aircraft engines are started
- All external servicing connections to the aircraft are removed
- The Nosewheel steering is set to ON
- The parking brake is set to OFF
- Get the applicable ground crew and/or ground control clearance to taxi

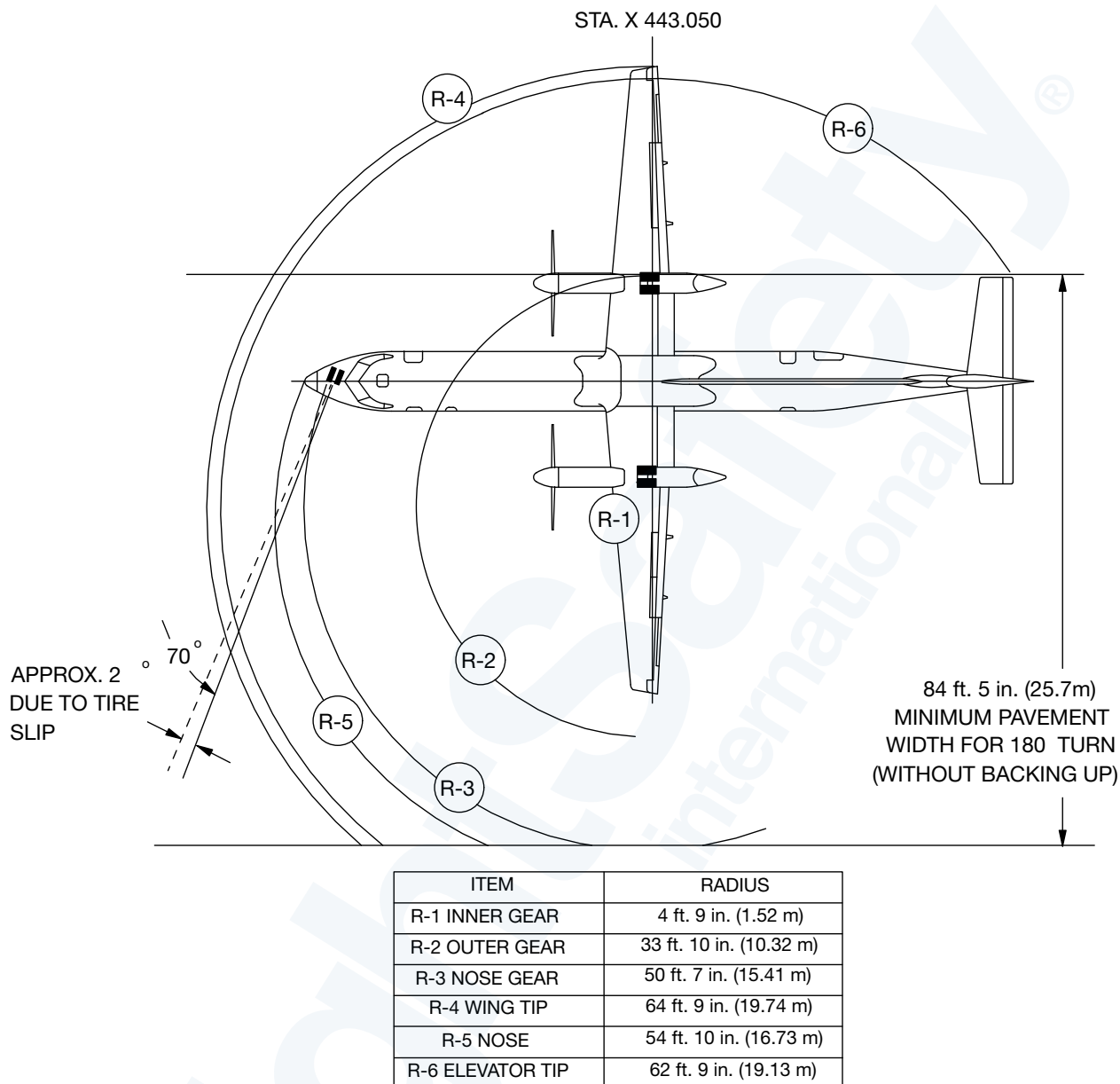
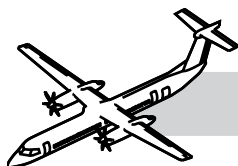
- With the condition levers set to MAX/1020 rpm, the POWER levers are used to apply sufficient engine power to start aircraft movement
- Pressure is applied to the brakes to make sure that the brake system is serviceable
- With the condition levers set to MAX/1020 rpm, the POWER levers are used to apply sufficient engine power to start aircraft movement
- The aircraft is taxied to the necessary area.

There are three taxi modes as follows:

- The high speed taxi mode gives a Nosewheel steering angle of 8° each side of center. Control the aircraft with the rudder pedals when you are in the high speed taxi mode
- The low speed taxi mode gives a Nosewheel steering angle of 70° each side of center. Control the aircraft with the Nosewheel steering hand-control on the pilot's side console
- The free-castering mode gives a Nosewheel steering angle of 120° each side of center. Control the aircraft with differential thrust and/or brakes.

When the taxi operation is completed, the engine power is decreased to the DISC position and the aircraft is stopped in a straight line.

Make sure that the Nosewheel is straight.

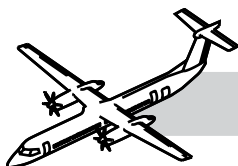


NOTES

1. Dimensions noted are for dry, hard, level surfaces at these tire pressures: 220 psi (loaded), 32 x 8.8-16 standard main-wheel tires, and 87 psi (loaded) for 22 x 6.50-10 standard nose-wheel tires.
2. Nose gear steering limit is approximately 70° left and right.
3. Slip angle of 2 is approximate only and may vary with aircraft configuration, loading and tire wear.
4. Dimensions given for maneuvering clearance and turning radii are minimum recommended limits.
5. Tire pressures shown are for calculation purposes only.

Refer to AMM Ch. 12 for service pressures.

Figure 9-5. Aircraft Taxiing Turning Radii with Minimum Power



09-00-00 SPECIAL TOOL & TEST EQUIPMENT

- GSB0900004 Towhead Assembly
- GSB0900005 Towbar
- None specified Tow Vehicle
- Lektro Model AP8750B-AL Tow Vehicle
- Lektro Model AP8850SDA Tow Vehicle
- Kalmar AB TBL 50 Tow Vehicle.

09-00-00 CAUTIONS & WARNINGS

CAUTION

Obey the precautions that follow when you tow the aircraft:

- Do not tow at more than 5 mph (8 kph)
- Do not use the emergency/parking brakes to stop the aircraft unless there is an emergency
- Make turns large when possible
- Make all changes to speeds or directions slowly
- When you stop the tow vehicle, do not make a sudden change to the opposite direction
- Turn limits must not be more than 120° right and left from the center.

CAUTION

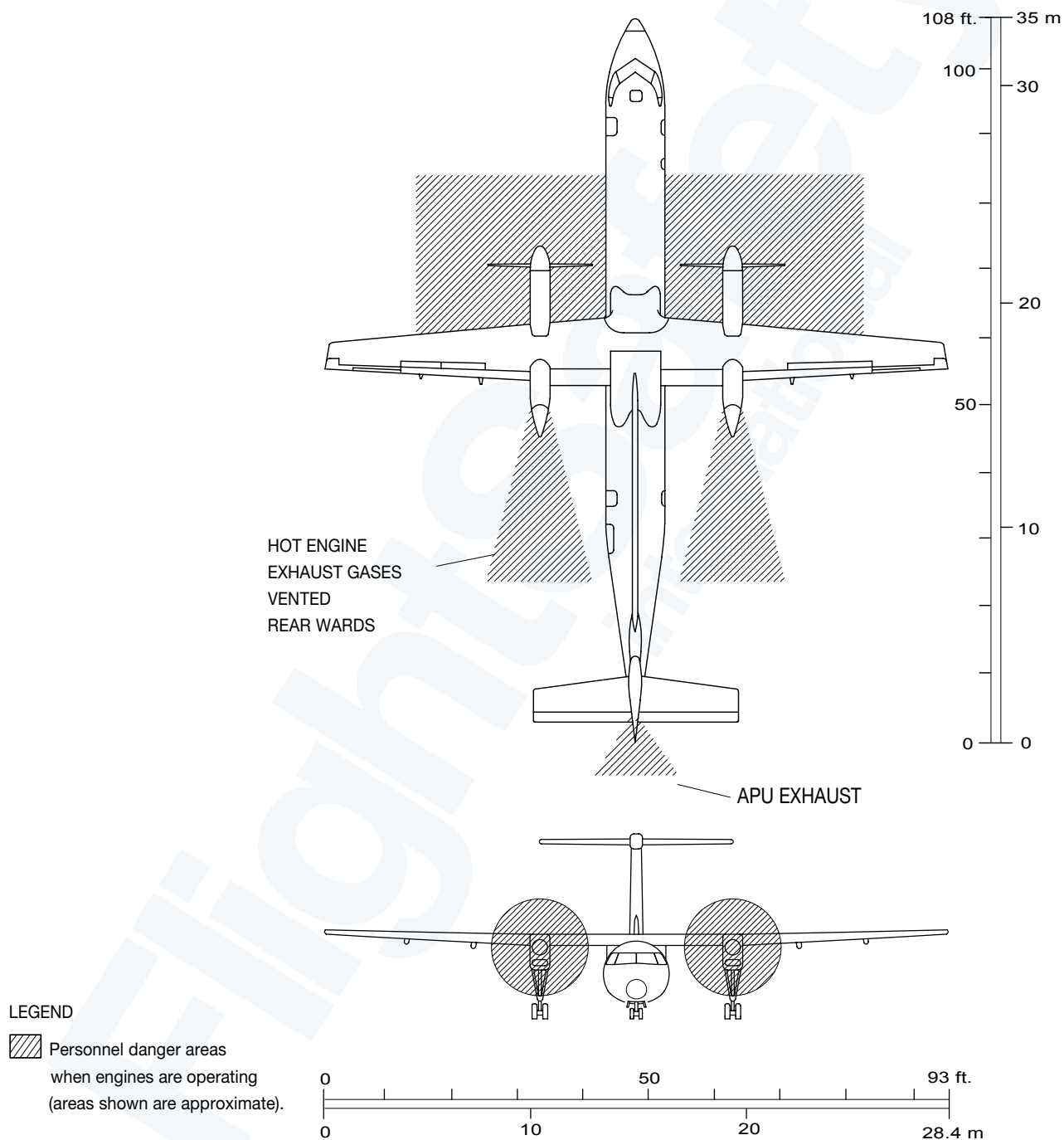
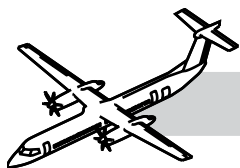
Do not tow at more than 3.5 MPH (5.6 KPH) for the model AP8750B-AL or 4.0 MPH (6.5 KPH) for the model AP8850SDA or 15.5 MPH (25 KPH) for the Kalmar Motor AB TBL 50.

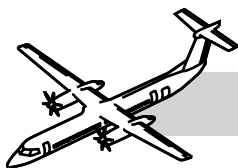
CAUTION

Do not stop and then start the tow vehicle with the nose landing gear at the 90° position.

CAUTION

Limit the applied torque to 10,000 lbf in (1130 Nm) or 7,500 lbf in (847 Nm) for the Kalmar Motor AB TBL 50.



**CAUTION**

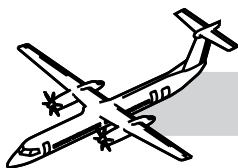
For the Kalmar Motor AB TBL 50, do not exceed the maximum fore-aft tow load of 5,513 lbs (24,532 N) tension and 5,884 lbs (26,180 N) compression.

CAUTION

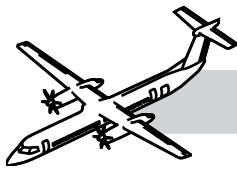
Make sure that you obey the instructions for tow vehicle operation given by the manufacturer of the tow vehicle.

CAUTION

Do not use power lever settings above flight idle or aft of disc when you are around areas of contamination on ground. If you do, you can cause the contamination to go into the engine intake and cause damage (Figure 9-6).



10-00-00 PARKING, MOORING AND RETURN TO SERVICE



10-11-00 PARKING

INTRODUCTION

While the aircraft is parked, safety is the prime concern. Bombardier has documented procedure for parking the aircraft that will be described in this chapter. Bombardier does understand that parking conditions may vary such as during a quick turnaround. During such events, the operators may choose to omit certain tasks at their discretion.

GENERAL

The steps that follow are done at all times when the aircraft is parked in or out of the hangar:

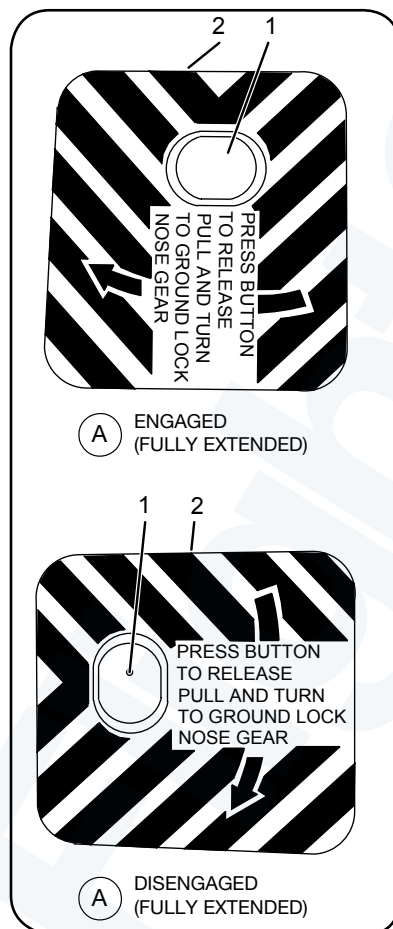
- The parking brake is set
- The ground lock for the nose landing gear is engaged
- The lockpins are installed in the main landing gear
- The control lock is set
- The aircraft is grounded
- The wheel chocks are installed at the nose landing gear and at the main landing gear.

Refer to Figure 10-1. Engaging the Nose Gear Ground Lock.

The nose gear ground lock is engaged as follows:

- The pushbutton is pushed in to release the door of the NLG ground lock
- The door of the nose gear ground lock is pulled away from the fuselage
- The door of the nose gear ground lock is turned 90 °s clockwise to engage the ground lock.

If the nose gear ground lock cannot be engaged correctly, install a jack adapter and put a jack below at the forward fuselage jacking point.

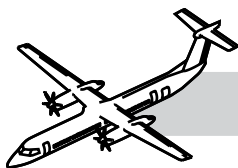


LEGEND

1. Release button

2. Nose gear ground lock door.

Figure 10-1. Engaging the Nose Gear Ground Lock



Refer to Figure 10-2. Installation of the Main Landing Gear (MLG) Lock Pins.

The MLG lockpins are installed in the lockpin holes of the stabilizer down-lock mechanism.

If the MLG lockpin cannot be put in its hole easily, install a jack adapter and put a jack below the applicable wing jacking point.

The jack can be extended sufficiently to remove the weight of the aircraft from the main landing gear.

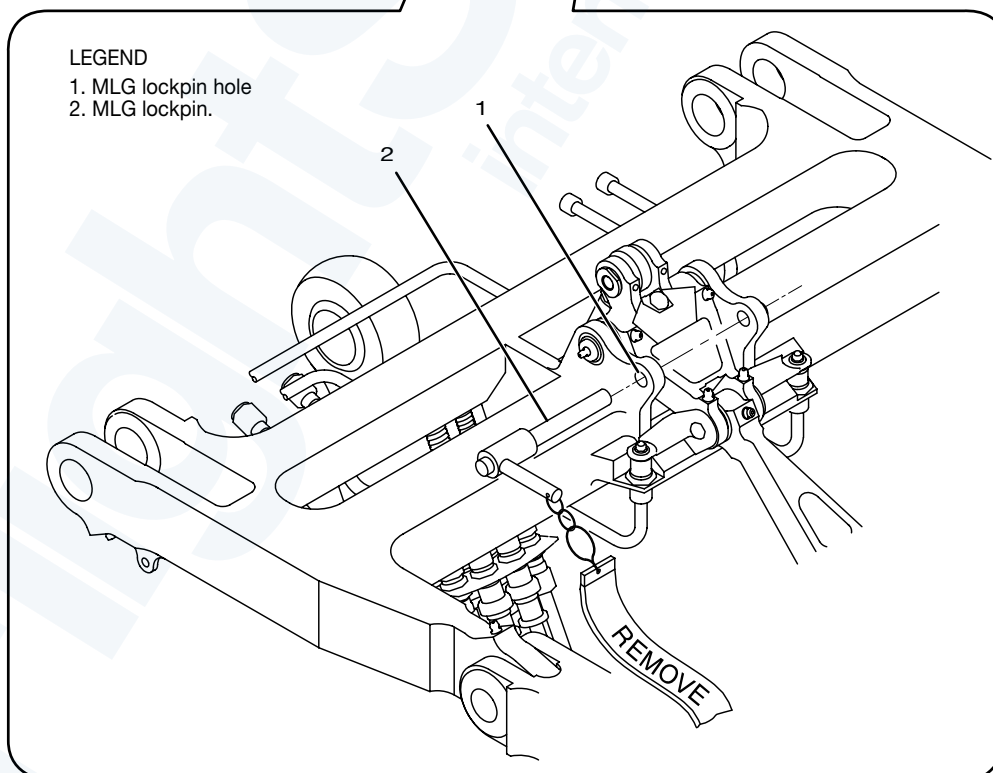
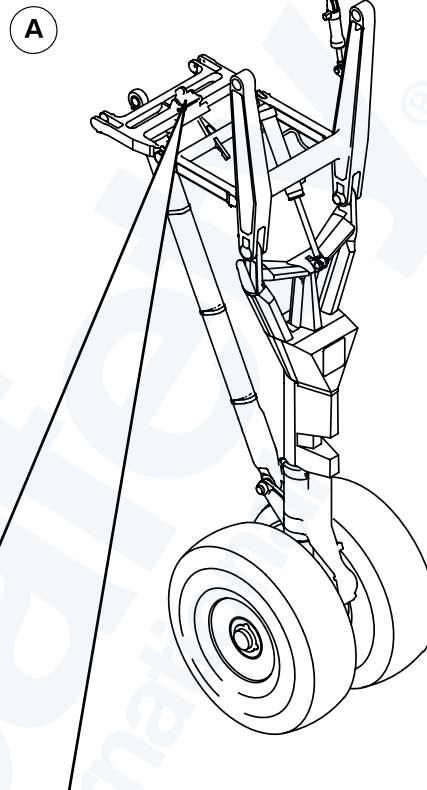
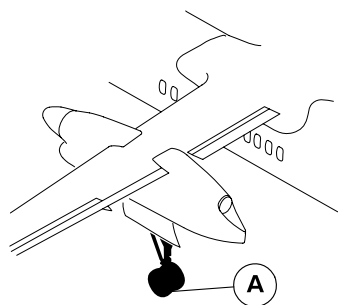


Figure 10-2. Installation of the Main Landing Gear (MLG) Lock Pins

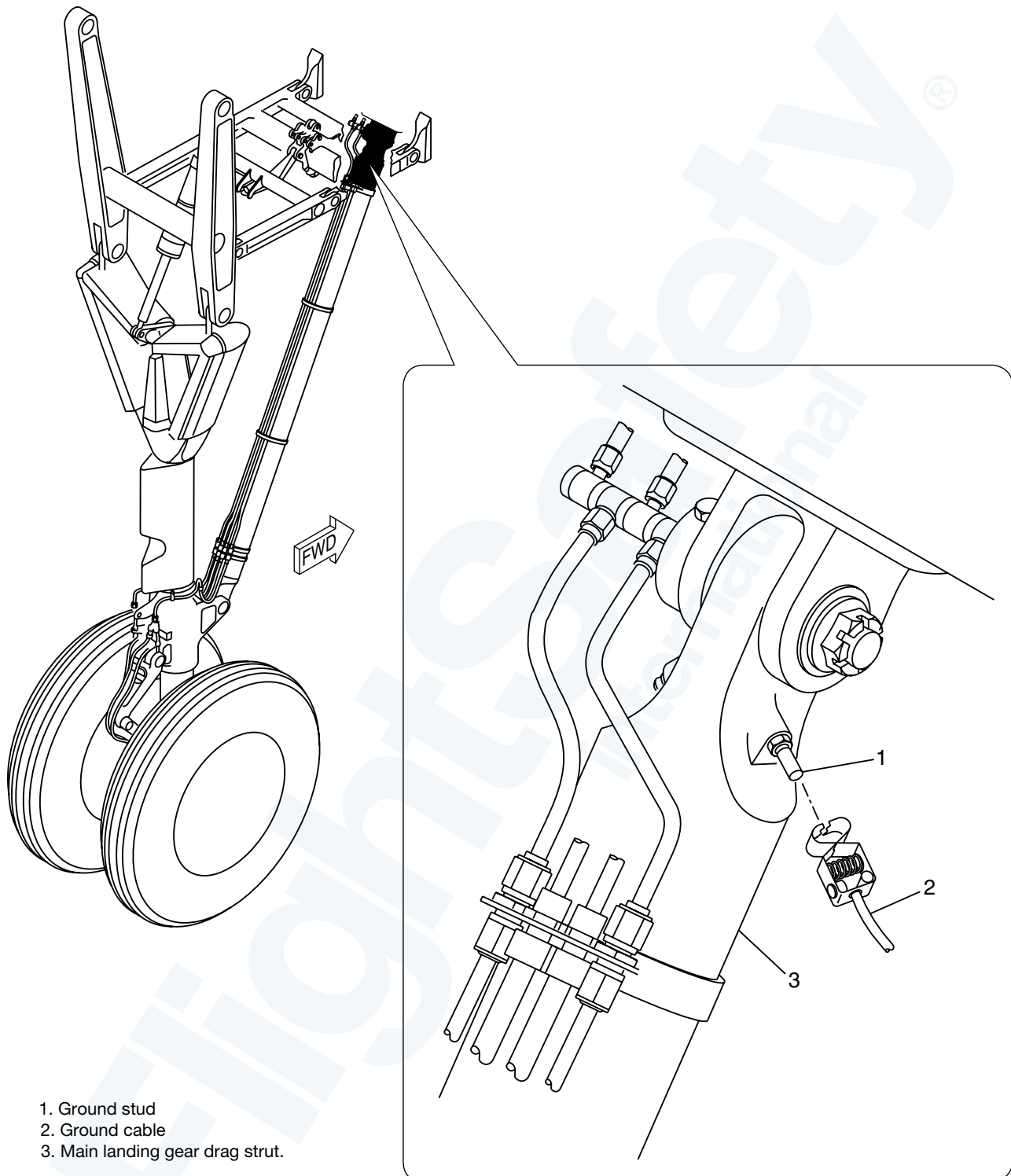
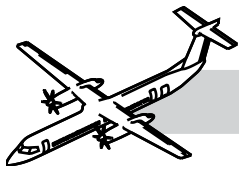
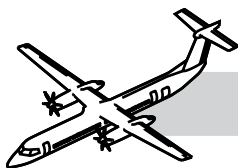


Figure 10-3. Static Grounding of the Aircraft



Refer to Figure 10-3. Static Grounding of the Aircraft.

NOTES

There is a ground point on the drag brace of each main landing-gear assembly. There is no ground point on the nose landing gear.

One end of the grounding cable is attached to a correct ground point in a hangar or the ramp area.

The alligator clamp of the grounding cable is attached to the ground stud on the applicable main landing gear drag brace.

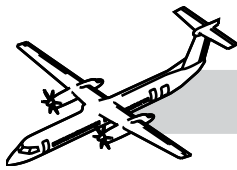
If for any reason the main landing gear is retracted, the aircraft may be grounded through these alternate grounding points:

- Pressure refuelling panel.
- Over the left or right wing.

If the MLG are retracted but the MLG door is open, the aircraft should be grounded through the ground stud.

When the aircraft is parked outdoors, the following steps should be done:

- The aircraft nose is pointed into the wind
- The engine intake and exhaust covers are installed
- The wheel chocks at the nose landing gear and at the main landing gear are installed
- The covers for the pitot-static probes are installed
- The covers for the angle-of-attack vanes are installed
- The propeller restraints are installed
- The wheel covers on the nose landing gear and the main landing gear are installed as necessary
- All other supplied covers and plugs are installed as necessary.



10-12-00 AIRCRAFT STORAGE

INTRODUCTION

An aircraft must be in an airworthy condition before it is put into storage. This means that hourly and calendar inspections must be done before the aircraft is put into storage. This does not apply if storage becomes necessary to complete extensive repairs. For example, if an aircraft has major damage to various components and repairs are estimated to take 14 days, scheduled tasks that become due during storage must be completed before the aircraft is returned to service.

GENERAL

Terms and References

AS NECESSARY: The step depends on local conditions. Do these steps at the discretion of the Director of Maintenance.

DESICCANT: Where desiccant is used, it must be put on a non-porous tray or wooden rack. This will make sure that the moisture absorbed by the desiccant will not cause corrosion on the structure or mildew on the fabric.

ENGINE OIL ANALYSIS: This is a report of the content of an oil sample taken from a specific engine. The report identifies various components of the oil sample. The total acid number (TAN) and percent water content are important for engines being prepared for storage.

An engine oil analysis is necessary to make sure that oil remaining in a stored engine does not have too much water or acid that could cause damage to the engine. Engine oil analysis laboratories are listed in the World Aviation Directory.

FLY-SCREEN COVERS: This term is used where it is intended that an orifice be covered to prevent insects or small birds from entering, but will allow ventilation of the area. Fly-screen covers are held in place with pressure sensitive tape. Use a tape color that is in contrast with the aircraft paint scheme.

When fly-screen cover is installed over recessed area, such as the emergency exit door handle, the screen shall cover the recessed area such that allows collected moisture to drain. The tape that holds the cover in place shall not overlap the recessed area.

Humidity Indicators

Humidity control is maintained through the use of desiccant in sealed compartments. Humidity level is shown by humidity indicator cards. These cards show a blue color when humidity is less than 40% and a pink color when humidity is more than 40%.

Engine Preservation

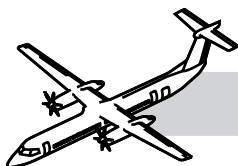
Aircraft that are stored for 28 consecutive days or less do not require engine preservation protection if the engine is sheltered, humidity is not too high, and the engine is not subject to extreme temperature changes. The engine must be operated on the ground every 7 days. Refer to the WEEKLY REPETITIVE TASKS table for details.

Environmental Considerations

The storage schedules listed in this procedure include steps for all types of environmental conditions.

The general intent of these storage schedules is to make sure that the aircraft is correctly;

- Preserved to readily return the aircraft to service
- Equipped to absorb and show the presence of excessive moisture
- Ventilated to prevent the collection of condensation
- Sealed against different types of precipitation, insects, and wildlife
- Cycled to give necessary maintenance for engines/propellers, hydraulic systems, flight controls, de-icing/anti-icing systems, etc.



The Director of Maintenance must decide if all steps are necessary based on local conditions.

The nacelles must be sealed to permit sufficient air flow and inspection, but stop insects or other wildlife which could go into the nacelle.

Insects can assemble nests in important drains. Birds can also assemble nests in the aircraft. This can cause structural deterioration and is also dangerous to personnel. Examine the aircraft for evidence of insects, birds, or other animal infestation.

Storage Records

The individual aircraft storage record must be kept current by the storage agency to show compliance. Send copies of the record to the airline's Director of Maintenance or his delegate.

The maintenance procedures that follow are given in a checklist form with space for the mechanic's initials (MECH INIT), an inspection/check stamp (I/C STAMP), and the date the STEP was done.

At the discretion of the airline's Director of Maintenance or his delegate, certain steps may be marked as not applicable (N/A) due to environmental considerations or operational requirements.

If a step is not applicable (N/A), the MECH INIT block should be marked N/A and the inspector should stamp the I/C STAMP block.

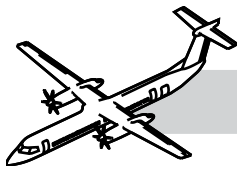
Certain materials listed in the DESCRIPTION can be substituted, at the discretion of the airline's Director of Maintenance, because of environmental considerations, operational requirements, or the local availability of the materials. Only equivalent materials may be used.

Additional procedures and materials may become necessary for the completion of certain checklist STEPS given in this procedure. It is not practical for this procedure to include all possible references.

Refer to the manuals that follow as necessary:

- Aircraft Maintenance Manual, PSM 1-84-2
- Fault Isolation Manual, PSM 1-84-5
- Maintenance Task Card Manual, PSM 1-84-7TC
- Consumable Products Manual, PSM 1-84-18

If you prepare the aircraft for storage for a maximum of 28 days and it is necessary to store it for a longer period of time, refer to the table STORAGE TRANSITION PROCEDURE FROM MAXIMUM 28 DAYS TO MAXIMUM 90 DAYS. This table gives the necessary additional instructions to extend the storage period to 90 days maximum.



10-13-00 ENGINE PRESERVATION/DEPRESERVATION

INTRODUCTION

Two procedures are given to preserve the engine. The first gives the preservation instructions for engines that are stored on the aircraft or stored off the aircraft or in a Quick Engine Change (QEC) unit for a time limit that is known. The second procedure gives the preservation instructions for engines that are stored on the aircraft for a time limit that is not known.

GENERAL

During engine preservation, all the openings to the engine are closed and sealed. The engine is put in a sheltered location that has a stable temperature and a low level of humidity.

If an engine is stored on the aircraft for 8 to 28 days:

- The following is done prior the engine is started and run until the oil temperature is a minimum of 70°C (158°F) for 10 minutes between GI and TO power or until the oil temperature becomes stable at TO power
- The engine is shut down
- A TAN oil analysis is done to find the acidity and calculate the water content of the oil
- If the TAN analysis shows oil acidity or water content, the engine oil is drained and new oil is put into the engine
- The engine is started, run and shutdown again
- A TAN analysis is done
- If the TAN analysis shows oil, or water content, by weight or by volume, the engine run and the oil change procedure is done again

- If the TAN analysis shows no oil acidity, or water content, there is no action required.

If the aircraft will be stored outside the following is done:

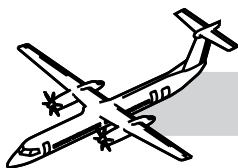
- All the openings to the engine are closed and sealed
- Every 7 days, the engine is started and run until the oil temperature is a minimum of 70°C (158°F) for 10 minutes between GI and TO power or until the oil temperature becomes stable at TO power.

If the engine will be stored in a hangar:

- Desiccant and a humidity indicator are put on racks made of wood in the engine inlet duct
- All the openings to the engine are closed and sealed
- While the engine is stored, the humidity indicator is viewed every seven days
- If the color changes to pink, it must be reactivated or replaced
- The visual inspections are recorded and attached to the engine.

If the engines are stored off the aircraft or in a QEC unit:

- Desiccant is put on wood racks in the air inlet and in the exhaust duct
- All the openings to the engine are closed and sealed
- The engine is put in a sheltered location that has a stable temperature and a low level of humidity to keep condensation to a minimum.



If the engines are stored for 29 to 90 days on or off the aircraft or in a QEC unit:

- A visual inspection of the humidity indicator is to be conducted every 7 days
- The oil system is preserved
- The fuel system preserved
- Desiccant and a humidity indicator are put on racks, made of wood, in the air inlet and in the exhaust duct
- All the openings to the engine are closed and sealed.

If the engines are stored for over 90 days, on or off the aircraft or in a QEC unit:

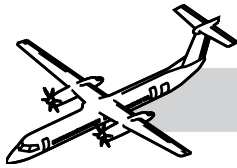
- A visual inspection of the humidity indicator is to be conducted every 7 days.
- The outside area of the engine is washed
- A visual inspection of the engine externals are done
- Anticorrosion fluid inhibitor is applied to the engine external surfaces (including the propeller shaft)
- The oil system is preserved
- The fuel system is preserved
- Desiccant and a humidity indicator are put on wood racks in the air inlet and in the exhaust duct
- All the openings to the engine are closed and sealed
- A spray of engine oil is applied to the accessory drive pads and shipping covers are installed.

If an engine is installed on the aircraft for a time limit that is not known, it is preserved as follows:

- The engine is started and run until the engine oil temperature is a minimum of 70°C (158°F)
- The engine is shut down
- A TAN oil analysis is done to find the acidity and calculate the water content of the oil
- If the TAN analysis shows acidity, or the water content; the engine is drained, new oil is put into the engine, the engine started and run and then the engine is shut down
- If the TAN analysis shows oil acidity or water, the engine run and oil change procedure is conducted again
- If the TAN analysis shows no oil acidity, or the water content, there is no action required.

The engine is depreserved as follows:

- Desiccant bags, shipping covers, caps and plugs are removed from the engine
- The aircraft fuel supply tube is connected to the FMU
- New oil is put into the engine.
- The fuel supply tube at the flow divider disconnected
- A container is put below the fuel supply tube
- A wet motoring run is done to see if a solid stream of fuel comes out of the fuel supply tube
- The condition lever is moved to FUEL OFF to see if the fuel flow stops
- The condition lever is moved to START/FEATHER to see if a solid stream of fuel comes out of the fuel supply tube
- The fuel supply tube is connected to the flow divider.



10-20-00 MOORING OF THE AIRCRAFT

INTRODUCTION

Mooring of the aircraft is done to secure the aircraft to the ground while the aircraft is in a parked position. Mooring is done during high wind conditions or when the aircraft is parked on snow or ice covered surfaces. Rope and ground attachment rings are used to moor the aircraft at the nose landing gear and at both the main landing gears.

Bombardier recommends that the aircraft must be moored after being placed in the parked condition outside or relocated to a protected location for the following conditions:

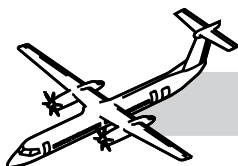
- When it is parked on dry concrete or asphalt and the winds are (or are forecast to be) 63 mph (102 km/h = 34.02 Knots) or more
- When it is parked on wet concrete or asphalt and the winds are (or are forecast to be) 46 mph (74 km/h = 40 knots) or more
- When it is parked on snow or ice covered surfaces.

Refer to Table 10-1. Wind Speed and Direction Limitations for Aircraft Mooring.

The wind speed and the wind direction limitations given in table 10-1 pertains to a moored aircraft regardless of ramp surface conditions.

Wind Speed (knots)	Wind Angle (knots)
65	90
66	80
68	70
71	60
74	50
78	70
83	40
88	30
95	25
110	15
130	10-0

Table 10-1. Wind Speed and Direction Limitations for Aircraft Mooring



GENERAL

NOTES

Refer to Figure 10-4. Mooring the Aircraft at the Nose Landing Gear.

The aircraft is moored at the nose landing gear as follows:

- A loop is made with the rope and the rope is put around the tow spools of the nosewheel and put through the ground attachment rings
- Attach the ropes to the ground attachment rings with tension
- A bungee cord is installed between the rope and the middle ground attachment ring to ensure that the rope has tension.

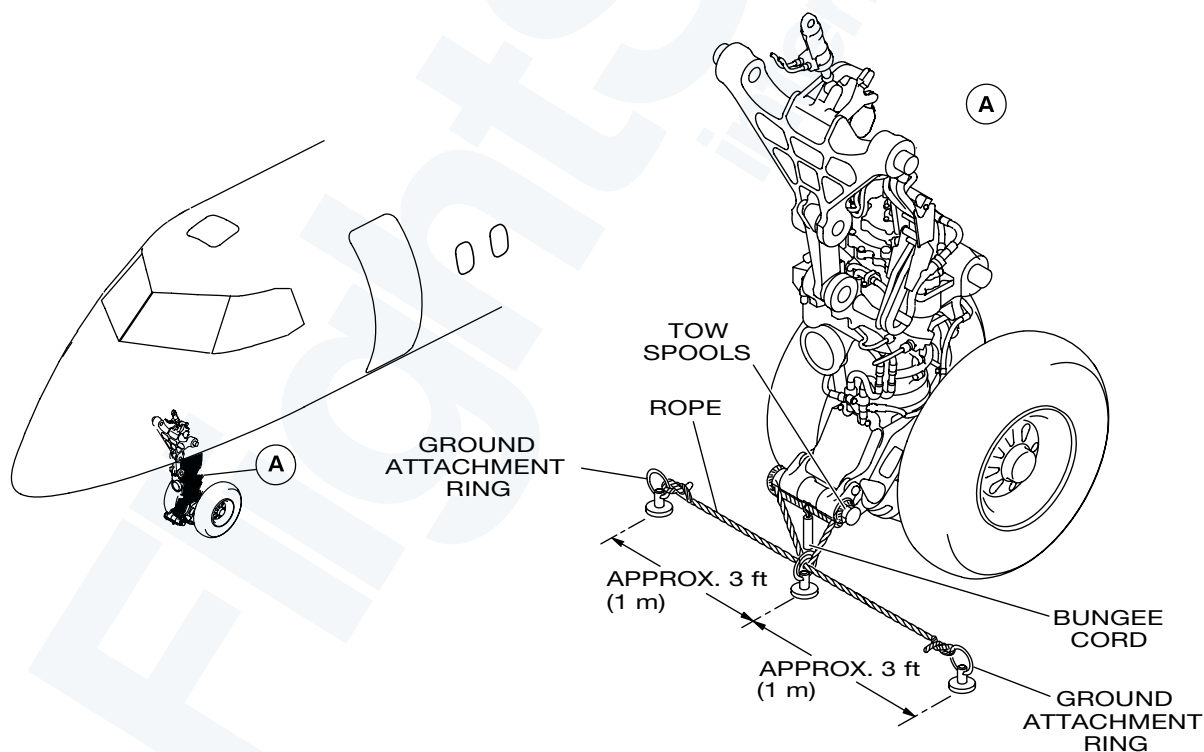
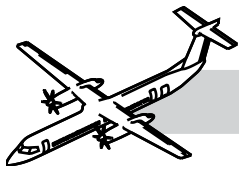


Figure 10-4. Mooring the Aircraft at the Nose Landing Gear



Refer to Figure 10-5. Mooring the Aircraft at the Main Landing Gears.

The aircraft is moored at the main landing gears as follows:

- A protective spool is installed at each main landing gear shock strut outer web

- A double loop is made with the rope around the protective spool on the outer web of the main landing gear and put through the attachment rings
- The ropes are attached to the ground attachment rings with tension.

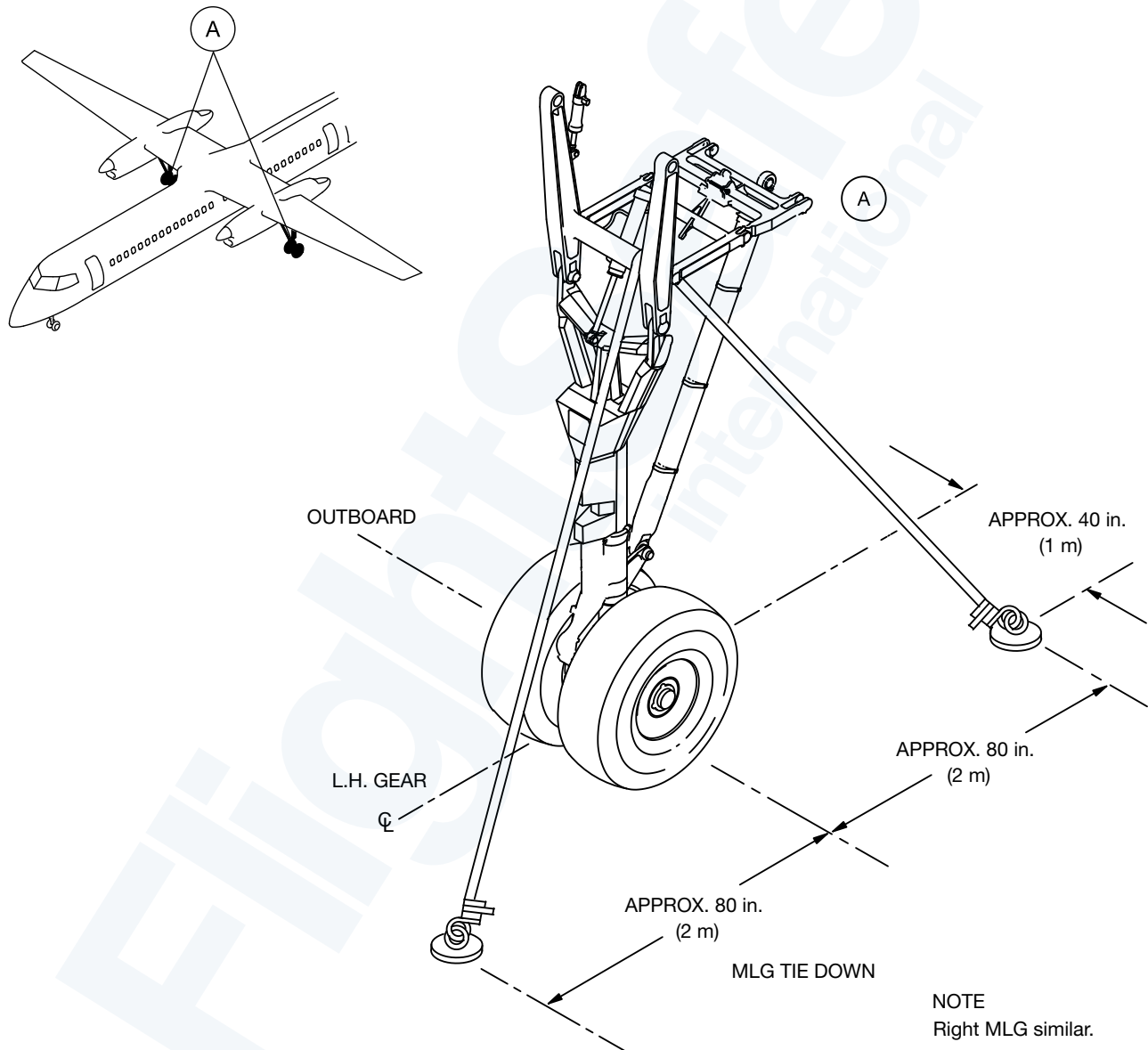
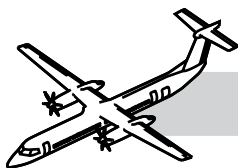
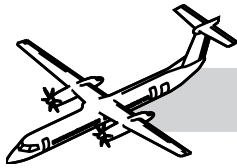


Figure 10-5. Mooring the Aircraft at the Main Landing Gears



10-00-00 SPECIAL TOOL & TEST EQUIPMENT

- 14-6882-6010 Covers, Main Wheels
- 14-6882-6000 Covers, Hi-flt MLG wheels
- 400-76200-13 Plug, APU Exhaust
- 400-76200-15 Cover, APU Intake
- AOA 100 Cover, Angle of Attack Vane
- ACS 28001 Cover, Engine Intake
- ACS 28002 Cover, Engine Exhaust
- ACS 28010 Plug, Oil Cooler Vent Scoop
- GSB1000004 Wheel Chocks - MLG and NLG
- GSB1000016 Cover, Nosewheel
- GSB1000033 Covers, Pitot Static
- GSB1000031 Propeller Restraint
- GSB1011007 Covers, ECS Exhaust
- GSB1011008 & GSB1011009 Covers, ECS RAM Air Duct
- GSB1011010 or equivalent protective spool
- GSB5411003 Cover, Engine Lower Cowl
- ACS28003 Cooler Intake Plug
- ACS28010 Plug, Oil Cooler Vent Scoop
- GSB0700016 Jack-Axle Nose Landing Gear
- GSB0700023 Jack-Axle Main Landing Gear
- GSB1000029 Pins, MLG Ground Lock (2)
- GSB1000037 Cover, APU
- None specified Humidity Indicator (Hygrometer)
- None specified Flashlight (non-explosive), with Batteries.
- Non specified Ropes, nylon or polypropylene 0.75 inch (19 mm) diameter minimum
- Non specified 7 ground attachment rings
- Non specified Bungee cord.



10-00-00 CAUTIONS & WARNINGS

WARNING

Do not install the covers when the pitot static probes are hot. The cover may melt and cause a blockage of the holes in the pitot static probe.

WARNING

Do not use force when you engage the ground lock of the nose landing gear. If force is necessary, the landing gear may not be safe and can retract accidentally.

WARNING

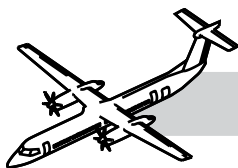
Do not use force when you install or remove the lockpin. If force is necessary, the landing gear may not be safe and can retract accidentally.

CAUTION

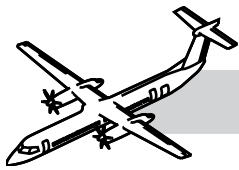
Do not put engine oil into the engine air inlet or exhaust. Rotor and stator components that are wet with oil can cause dirt particles to attach to their surfaces. This can change the airfoil shapes and have an effect on the efficiency of the engine.

NOTE

To reduce any pitching tendencies during high winds, it is recommended that the aircraft flaps are set to 0 °s, the aircraft be fully fueled and the center of gravity be moved as far forward as possible.



11-00-00 PLACARDS AND MARKINGS



11-00-00 PLACARDS AND MARKINGS

INTRODUCTION

The aircraft has placards, labels, and markings.

The aircraft has exterior placards and markings on the following:

- Fuselage
- Landing Gear
- Wing
- Nacelles and Propellers.

The aircraft also has interior placards and markings on the following:

- Flight Compartment
- Lavatory
- Cabin
- Galley
- Baggage Compartments.

GENERAL

Refer to Figure 11-1. Left Fuselage Placards and Markings (Example Configuration).

Some placards are mandatory because of local airworthiness regulations. Please refer to your local regulation authority requirements.

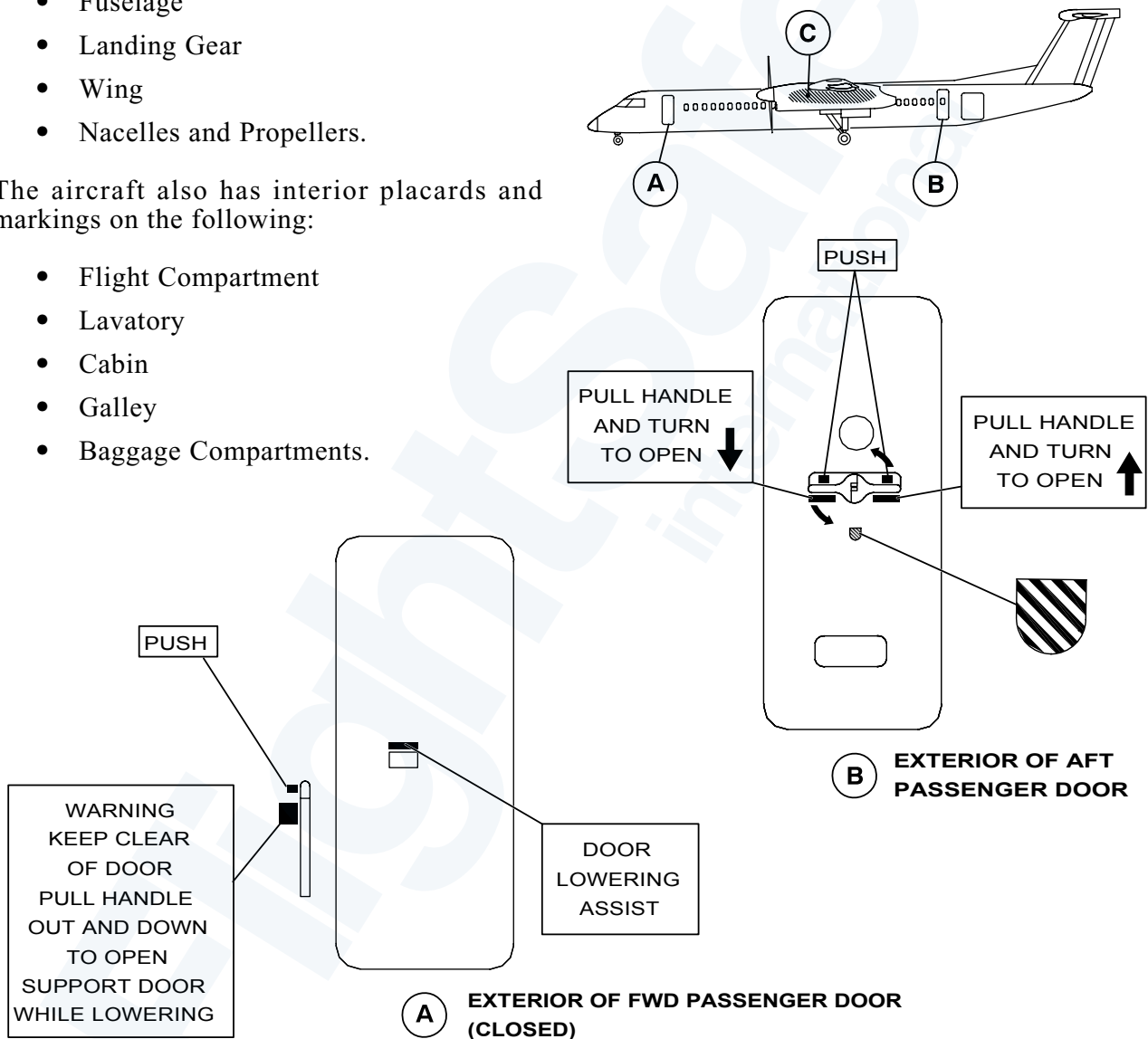


Figure 11-1. Left Fuselage Placards and Markings (Example Configuration)